IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 2 September 2025, accepted 20 September 2025,
date of publication 24 September 2025, date of current version 30 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3614058

RESEARCH ARTICLE

An Innovative Hybrid Machine Learning Approach for Student Survey Analysis: Random Tree With Ordinal Noise Filtering and Feature Selection (RTONF)

GOKSU TUYSUZOGLU[1], YUNUS DOGAN[1], FERISTAH DALKILIC[1], ELIFE OZTURK KIYAK[2], BITA GHASEMKHANI[3], KOKTEN ULAS BIRANT[1,4], AND DERYA BIRANT[1], (Member, IEEE)

[1]Department of Computer Engineering, Dokuz Eylul University, 35390 İzmir, Türkiye
[2]Independent Researcher, 35140 İzmir, Türkiye
[3]Graduate School of Natural and Applied Sciences, Dokuz Eylul University, 35390 İzmir, Türkiye
[4]Information Technologies Research and Application Center (DEBTAM), Dokuz Eylul University, 35390 İzmir, Türkiye

Corresponding author: Derya Birant (derya@cs.deu.edu.tr)

ABSTRACT Analyzing student survey data using machine learning has become increasingly important for educational institutions aiming to understand the student experience, identify areas for improvement in teaching and curriculum, and make informed decisions to enhance learning outcomes. A major challenge in this domain is the presence of noisy data, which can substantially reduce the performance of classification algorithms. Existing studies often ignore the ordinal nature of class labels during noise detection, which may lead to suboptimal data cleaning. To address this problem, we propose a new approach entitled Random Tree with Ordinal Noise Filtering and Feature Selection (RTONF). This method explicitly incorporates the inherent order of class labels (e.g., poor < fair < good < very good < excellent) during the noise identification process, before predicting student performance or satisfaction. The Random Tree (RT) algorithm serves as the base classifier, while Pearson Correlation is employed for feature selection due to its outstanding performance. Experimental results demonstrated that the proposed hybrid method with an average accuracy of 84.61% achieved a 5.23% improvement compared to the traditional RT classifier. Furthermore, comparative analysis indicated that our method outperformed the state-of-the-art techniques in terms of prediction accuracy on the same dataset.

INDEX TERMS Artificial intelligence, classification, education, educational data mining, feature selection, machine learning, noise detection, student performance prediction, student satisfaction prediction, survey data.

I. INTRODUCTION

In the educational data mining, analyzing student survey data to predict student performance or satisfaction is crucial because it can provide actionable understandings that support educators in boosting learning outcomes and improving overall student attendance, motivation, and academic success. Identifying the primary factors affecting student performance/satisfaction prediction can help educational administrators implement more effective strategies for enhancing educational quality. Therefore, it's essential to understand the key teaching-related aspects that impact student performance/satisfaction in order to make informed decisions. Recent studies have shown that survey-driven predictive analytics can reveal latent relationships between teaching effectiveness, course content, and student engagement, leading to targeted improvements in curriculum design and instructional methods [1].

Focusing on educational data mining, the use of data-driven methods in educational research has significantly

The associate editor coordinating the review of this manuscript and approving it for publication was Aysegul Ucar.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

increased, particularly in the analysis of student feedback and institutional effectiveness. Machine learning (ML) approaches have become powerful tools for capturing complex interactions among various pedagogical and behavioral variables. An ML model can reveal relationships among instructional performance, course constructs, learning activities, student success, and student satisfaction, allowing instructors to adjust their teaching methods to better meet student expectations. It can also help educators enhance decision-making on improving the learning process by uncovering hidden patterns within educational datasets [2]. Accordingly, our study aims to develop ML models capable of accurately predicting student performance and satisfaction from survey data based on their perceptions of course components and instructional performance.

The analysis of student evaluations collected from surveys is a challenging issue, as student performance and satisfaction depend on various aspects such as course content, the number of course repeats, class attendance, perceived difficulty level of the course, and instructor behavior. The complexity of student survey data, including subjective feedback and diverse course content, introduces high computational overhead for learning algorithms and increases the risk of overfitting, thereby making it difficult to capture non-linear patterns and reducing prediction accuracy. The widespread use of information technology systems in education produces vast amounts of data, posing challenges to analyze and uncover meaningful insights. A large number of data instances provides rich information, but it also increases uncertainty and noise. Noisy data is a common difficulty that can substantially degrade the performance of classification algorithms, making data cleaning a critical preprocessing step [3]. To overcome this challenge, our study aims to develop a noise-filtering method to enhance the quality of educational data.

The main problem is that traditional noise identification methods are nominal in nature and don't consider the inherent ordering of class labels in the student survey data (e.g., poor < fair < good < very good < excellent). Actually, the class labels follow a natural order—for example, an instance labeled ''good'' is rated higher than one labeled ''poor'', while an instance labeled ''excellent'' is better than both. However, conventional noise detection techniques often treat these labels as unordered categories, potentially misclassifying minor deviations in ratings as significant errors. Therefore, a rank-oriented noise removal method is required, and as a new approach, it needs to be investigated thoroughly. By incorporating the ordinal nature of class labels, such methods can better preserve the underlying semantics of the data and improve the reliability of downstream predictions. This perspective aligns with the need for specialized preprocessing techniques that respect the ordinal properties of educational survey responses [4].

After the ordinal noise removal and feature selection process, this study applies the random tree (RT) [5] classification algorithm to the clean dataset to predict student achievement and satisfaction, aiming to help educators improve the quality of the teaching process. We developed machine learning models from self-reported survey data to frame student performance/satisfaction estimation as classification problems, aiming to predict different levels of student evaluations and uncover the influential factors regarding instructors and courses. The random tree algorithm, which builds a decision tree using a randomly selected subset of features at each node, is particularly well-suited for educational data mining tasks. Its advantages include robustness to overfitting and high interpretability, which is essential for ensuring that the model's logic can be reviewed and validated by practitioners in the education field. These properties are especially beneficial when handling nonlinear and heterogeneous variables found in educational datasets. The main contributions of this work are summarized as follows:

i. This study introduces the ordinal scale paradigm to the noise identification task. In this context, the education community has not yet investigated this paradigm.
ii. For the first time, it proposes a hybrid approach, random tree with ordinal noise filtering and feature selection (RTONF), to build robust machine learning models by leveraging the strengths of order-aware noise removal and feature selection.
iii. This study addresses two distinct but related tasks, including predicting student performance and predicting student satisfaction from the real-world student evaluation dataset. In contrast, most previous works focused on only one of these aspects. It explores the data related to student evaluations of courses and instructors to determine the factors that affect student performance and to infer student satisfaction predictors.
iv. Unlike some previous works, we developed an interpretable machine learning model that promotes transparency via random tree as the core classifier, and the proposed method achieved an average accuracy of 84.61%, representing a 5.23% improvement over the traditional random tree classifier.
v. The comparative study revealed that our method proved superior in predictive accuracy over the state-of-the-art methods on the same dataset.

The remainder of this paper is organized as follows: Section II presents a review of related work, focusing on applying machine learning techniques to student data in predictive modeling. Section III details the proposed RTONF method, explaining each of its components. Section IV presents the experimental setup. Section V demonstrates results, emphasizing the robustness of our proposed method. Section VI provides a comparative analysis with existing state-of-the-art methods. Finally, Section VII summarizes the conclusions and suggests directions for future research.

II. RELATED WORK

In recent years, several studies have investigated student survey data using machine learning techniques for a wide range of educational purposes [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42]. These studies vary across several dimensions, including the country of origin, specific courses examined, the type of datasets used (e.g., academic, behavioral, or survey-based), the machine learning methods applied, and the target class (binary, multi-class, or continuous). Additionally, the research differs in terms of task types—classification (C) and regression (R), the learning environments considered as online courses (O), universities (U), or high schools (H), the educational outcomes predicted such as performance, dropout, satisfaction, and the metrics used for evaluation, represented in Table 1.

The previous studies were conducted in a diverse set of countries, including China [6], [18], [34], [38], the UK [8], [22], [26], [35], the USA [16], [29], Canada [9], [23], Australia [10], [24], India [11], Brazil [12], France [13], Japan [14], South Korea [15], Peru [17], Somalia [19], Kuwait [27], Hungary [31], Germany [25], Italy [33], Jordan [36], Slovakia [37], Saudi Arabia [39], Bangladesh [40], Spain [41], and Hong Kong [42], as well as multi-countries [7], [20], [21], [28], [30], [32]. The studies covered a wide range of course domains. Some focused on specific disciplines such as Engineering [8], [12], [16], [40], Computer Science [14], Robotics [10], Business [23], Math [30], [33], and Programming [39]. Others addressed general or various courses across disciplines [6], [7], [9], [11], [13], [15], [17], [19], [21], [22], [25], [26], [27], [28], [29], [31], [32], [33], [34], [35], [36], [37], [38], [41], [42].

Furthermore, the studies utilized diverse data sources to analyze student performance and behavior. Survey data was the most commonly used source [6], [7], [17], [18], [19], [27], [31], [32], [35], [36], [37], [42], often capturing students' self-reported responses on attitudes, satisfaction, and academic experiences. Behavioral data, including log files and activity records, were also frequently employed [9], [10], [11], [13], [14], [16], [23], [25], [33], [34], reflecting students' engagement and learning patterns. Some studies relied on academic records such as grades and exam results [8], [12], [15], [22], [24], [29], [39], [41], while others used multiple data sources to combine in-sights from academic, behavioral, and survey inputs [20], [21], [28], [40]. A few works incorporated standardized datasets like PISA [26], [30].

A broad range of machine learning methods has been applied to analyze student data, reflecting both traditional and advanced techniques. Decision tree (DT) was among the most frequently used algorithms [8], [12], [15], [17], [19], [20], [21], [22], [28], [29], [31], [32], [34], [37], [39], [40]. Support vector machines (SVM) were also widely adopted for classification tasks [9], [11], [13], [16], [18], [19], [20], [21], [23], [28], [30], [31], [33], [34], [39], [40], [41], often in combination with other classifiers. The random forest (RF) algorithm has also been commonly used in prior studies [8], [10], [11], [13], [15], [19], [20], [21], [23], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [41], [42] focused on student data. Other ensemble algorithms, such as extreme gradient boosting (XGB), gradient boosting machines (GBM), and AdaBoost, were employed to improve predictive performance [6], [14], [18], [20], [21], [25], [28], [32]. Neural networks, including long short-term memory (LSTM) and artificial neural networks (ANN), were applied in several studies to model complex patterns [10], [20], [21], [27], [28], [36], [38], [40], [42]. Other techniques such as naive Bayes (NB), k-nearest neighbors (KNN), logistic regression (LR), and specialized methods like tree augmented naive Bayes (TAN), convolutional neural network-convolutional block attention module (CNN-CBAM), simple linear regression (RR), and ElasticNet were also used in various contexts [6], [7], [8], [12], [17], [19], [20], [21], [23], [24], [28], [31], [32], [33], [35], [36], [39], [40], [41], [42].

The prior studies addressed both classification and regression tasks, depending on the nature of the educational outcomes. The majority focused on classification tasks, predicting discrete categories such as student satisfaction, dropout status, pass/fail results, and grade categories [6], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [28], [29], [30], [31], [32], [33], [34], [35], [36], [38], [39], [40], [41]. Several studies addressed regression problems that the aim was to predict continuous outcomes like performance scores, grade point average (GPA), or satisfaction levels [7], [18], [19], [20], [21], [26], [27], [36], [37], [42]. Some works applied both classification and regression depending on the target variable or combined tasks within the same study [20], [21], [28].

Some studies explored student survey data collected from various learning environments, including online courses, universities, and high schools. A substantial number of studies focused on data from universities [6], [8], [10], [12], [15], [17], [22], [24], [28], [29], [33], [35], [38], [39], [40], [41], [42] while some studies analyzed data from high schools, capturing secondary education settings [9], [11], [13], [14], [16], [19], [20], [21], [23], [25], [26], [27], [31], [32], [34], [36], [37]. A smaller portion of research focused on online courses, reflecting the growing importance of digital learning platforms [7], [20], [21], [26], [27], [28], [31], [42]. Several studies included data from multiple environments, allowing for cross-contextual analysis [20], [21], [28].

Student survey data has been analyzed for various educational purposes, including predicting student performance [7], [18], [19], [29], [30], [31], [36], detecting students at risk of dropout [9], [13], [23], [41], estimating student satisfaction [6], [17], [26], [27], [35], [42], assessing student engagement level [34], projecting academic outcomes as pass/fail [8], [12], [15], [20], [21], [22], [25], [28], [38], [39], anticipating GPA scores [20], [21], [28], and classifying stress levels [32]. These diverse targets reflect the broad utility of

TABLE 1. Related works for student survey data analysis.

Ref	Year	Country	Course	Dataset	Methods	Target Class	C	R	O	U	H	Output	Metrics
[6]	2025	China	Various	Survey	XGB-FS, TAN	Binary	√			√		Satisfaction	AUC
[7]	2025	Various	Various	Survey	ML	Continuous		√		√		Performance	RMSE, MAE
[8]	2025	UK	Engineering	Academic	RF, DT	Binary	√			√		Pass/Fail	ACC, P, R, F1
[9]	2025	Canada	Various	Behavioral	SVM, NB	Binary	√		√	√		Dropout	ACC, P, R, F1
[10]	2025	Australia	Robotics	Behavioral	LSTM, RF	Binary	√		√		√	Grade	ACC, P, R, F1
[11]	2025	India	Various	Behavioral	SVM, RF	Binary	√		√	√		Grade	ACC, P, R, F1
[12]	2025	Brazil	Engineering	Academic	NB, DT	Binary	√			√		Pass/Fail	ACC, P, R, F1
[13]	2025	France	Various	Behavioral	SVM, RF	Binary	√			√		Dropout	ACC, P, R, F1
[14]	2025	Japan	Computer Science	Behavioral	XGB, KNN	Binary	√		√	√		Grade	ACC, P, R, F1
[15]	2025	South Korea	Various	Academic	RF, DT	Binary	√			√		Pass/Fail	ACC, P, R, F1
[16]	2025	USA	Engineering	Behavioral	SVM, NB	Binary	√		√	√		Grade	ACC, P, R, F1
[17]	2024	Peru	Various	Survey	DT, LR	Binary	√			√		Satisfaction	ACC, P, R, F1
[18]	2024	China	Mechanical Engineering	Survey	SVM, CNN-CBAM, XGB, CART	Multi-class	√			√		Performance	ACC
[19]	2024	Somalia	Various	NEAS	LR, RR, NB, DT, RF, SVM, KNN	Binary	√				√	Performance	ACC, P, R, F1
[20]	2024	Various	Introductory Programming	Multiple	DT, RF, SVM, ANN, LR, KNN, XGB, GBM	Binary Continuous	√	√		√		GPA, Pass/Fail	ACC, P, R, F1, AUC, RMSE, MAE
[21]	2024	Various	Various	Multiple	DT, RF, SVM, ANN, NB, LR, KNN, XGB, GBM	Binary Continuous	√	√		√	√	GPA, Pass/Fail	ACC, P, R, F1, AUC, RMSE, MAE
[22]	2024	UK	Various	Academic	KNN, DT	Binary	√			√		Pass/Fail	ACC, P, R, F1
[23]	2024	Canada	Business	Behavioral	RF, SVM	Binary	√		√	√		Dropout	ACC, P, R, F1
[24]	2024	Australia	Science	Academic	NB, KNN	Binary	√			√		Grade	ACC, P, R, F1
[25]	2024	Germany	Various	Behavioral	XGB, RF	Binary	√		√	√		Pass/Fail	ACC, P, R, F1
[26]	2024	UK, Japan	Various	PISA	RF, KNN	Continuous		√	√		√	Satisfaction	MAE, RMSE, R^2
[27]	2024	Kuwait	Various	Survey	ANN, RF	Continuous		√	√	√		Emotion, Satisfaction	MAE, RMSE
[28]	2023	Various	Various	Multiple	DT, RF, SVM, ANN, NB, LR, KNN, XGB, GBM	Binary Continuous	√	√	√	√	√	GPA, Pass/Fail	ACC, P, R, F1, AUC, RMSE, MAE
[29]	2023	USA	Various	Academic	DT, RF	Binary	√			√		Performance	ACC, AUC
[30]	2023	Asia-Pacific	Math	PISA	RF, LR, SVM	Binary	√				√	Performance	ACC, AUC
[31]	2023	Hungary	Various	Survey	SVM, DT, RF, KNN	Binary	√		√	√		Performance	ACC, F1
[32]	2023	Various	Various	Survey	DT, RF, AdaBoost, GB	Multi-class	√		√	√		Stress Levels	ACC, P, R, F1, ROC
[33]	2023	Italy	Math	Behavioral	SVM, RF, NB	Binary	√		√	√		Grade	ACC, P, R, F1
[34]	2023	China	Various	Behavioral	SVM, RF, DT, KNN	Binary	√			√		Engagement Level	ACC, P, R, F1
[35]	2022	UK	Various	Survey	RF, SVM	Binary	√			√		Satisfaction	ACC, P, R, F1
[36]	2022	Jordan	Various	Survey	ANN, SMO, REPTree, RF, SEM, CFA	Continuous		√		√	√	Performance	R^2, MSE
[37]	2022	Slovakia	Various	Survey	LR, DT, RF	Continuous		√		√		Grade	MAPE

TABLE 1. *(Continued.)* **Related works for student survey data analysis.**

[38]	2022	China	Various	LMS log	SVM, RF, ANN	Binary	√		√	√	Pass/Fail	ACC, F1-score
[39]	2021	Saudi Arabia	Programming	Academic	DT, kNN, NB, SVM	Binary	√			√	Pass/Fail	ACC, P, R, F1
[40]	2021	Banglade sh	Engineering	Academic & Survey	DT, NB, kNN, SVM, ANN, LR	Multi-class	√			√	Grade	ACC, Cost Complexity
[41]	2021	Spain	Various	Academic	SVM, RF, NB, KNN	Binary	√			√	Dropout	ACC, P, R, F1
[42]	2021	Hong Kong	Various	Survey	ElasticNet, MLP, KNN, SVR, RF, LGBM	Continuous		√	√	√	Satisfaction	MAE, RMSE, R^2

AUC = Area under the curve, CBAM = Convolutional block attention module, CNN = Convolutional neural network, DT = Decision tree, GBM = Gradient boosting machines, GPA = Grade point average, KNN = K-nearest neighbors, LR = Logistic regression, MAE = Mean absolute error, MAPE = Mean absolute percentage error, MLP = Multi-layer perceptron, NB = Naive Bayes, REPTree = Reduced error pruning tree, RF = Random forest, RMSE = Root mean squared error, ROC = Receiver operating characteristic, SVM = Support vector machine, TAN = Tree augmented naive bayes, XGB = Extreme gradient boosting

historical data for enhancing student support and academic success.

A variety of performance metrics were employed to evaluate the predictive models, with metrics chosen according to the task type. For classification tasks, common evaluation metrics included accuracy (ACC), precision (P), recall (R), and F-measure (F1), which were widely used across many studies [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [19], [20], [21], [22], [23], [24], [25], [28], [29], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41].

For regression tasks, metrics such as mean absolute error (MAE), root mean squared error (RMSE), R-squared (R2), and mean absolute percentage error (MAPE) were used to measure prediction accuracy and error [7], [18], [19], [20], [21], [26], [27], [36], [37], [42]. Additionally, some studies utilized metrics like receiver operating characteristic (ROC) and area under the curve (AUC) to assess classifier performance [6], [17], [20], [28], [32], [35].

The previous studies have demonstrated that the random tree algorithm outperformed other classifiers such as SVM, KNN, ANN, and NB in a variety of application domains [43], [44], [45], [46], [47], [48]. For instance, the random tree achieved the most accurate predictions of surface roughness in FDM-manufactured parts, outperforming C4.5 and RF [43]. Similarly, another study showed that the random tree attained the lowest RMSE and highest NSE in predicting the silt removal efficiency of vortex tube ejectors, surpassing SVM, RF, and MARS [44]. In the context of fuel property prediction, it was found that the random tree provided significantly lower prediction errors and near-perfect evaluations compared to ANN [45]. For hydrological modeling, it was reported that the random tree outperformed M5P, REPTree, and other tree-based methods in estimating daily river discharge [46]. In road accident analysis, it was demonstrated that the random tree achieved the highest classification accuracy (90.6%) among several decision tree algorithms [47]. Furthermore, the effectiveness of a random tree was emphasized in predicting students' academic performance within an intelligent recommendation framework [48]. These consistent results across diverse domains motivated our selection of a random tree as the base classifier in the proposed RTONF method.

Recent advances in ordinal noise handling have emphasized the need for methods that explicitly account for the structure of ordered categories. For instance, the inter-class distance-based filtering (ICDF) algorithm proposed in [49] relies on theoretical error bounds and refined distance measures to address the non-uniformity of ordinal labels. Similarly, diagnostics based on generalized residuals were introduced in [50] to identify anomalous and influential observations, demonstrating how model reliability can be compromised by atypical responses or covariate structures. Another relevant contribution was presented in [51] in the form of an optimal sample selection (OSS) framework, which utilizes generalization error bounds to guide the removal of noisy outputs in regression and ordinal classification. The framework also introduces the covering distance filter (CDF) to systematically separate low-noise samples from high-noise ones. Compared with these methods, RTONF stands apart by combining ordinally aware noise filtering with feature selection in a unified workflow. Rather than treating sample selection and feature relevance separately, it simultaneously enhances data quality and predictive capability.

Overall, predictive models predominantly rely on nominal noise identification approaches that ignore the inherent ordering in class labels (e.g., poor < fair < good < very good < excellent), which can negatively affect classification performance in ordinal datasets. In other words, existing studies typically treat labels as purely categorical, without accounting for their ranked structure during the noise detection phase. To overcome this limitation, we propose a novel method, RTONF, specifically designed to detect noisy instances while preserving the ordered nature of the class labels.

III. PROPOSED METHOD: RTONF

This study aims to develop ML models from student survey data to predict student performance and satisfaction. Analyzing such data using ML is particularly valuable, as it allows educational institutions to proactively identify and address factors that may impact student engagement,

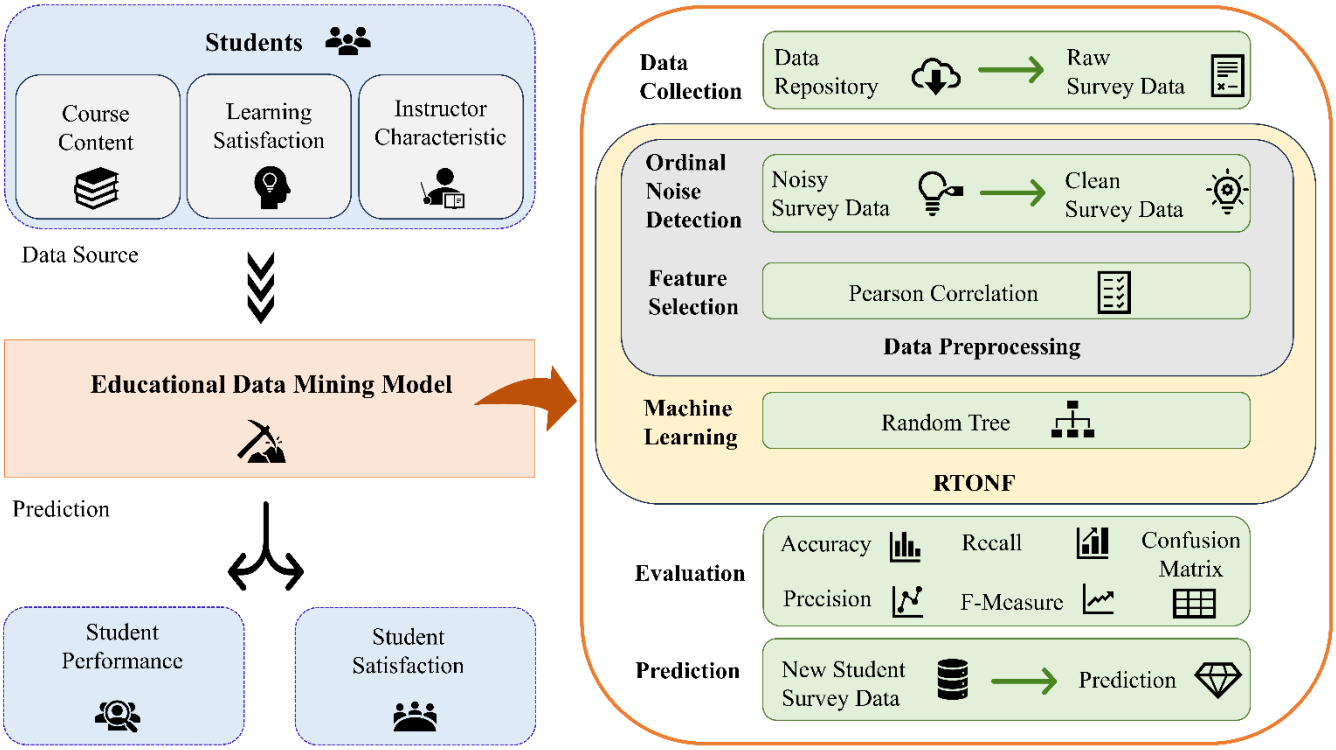


FIGURE 1. General architecture of the proposed RTONF method.

academic performance, and retention, well before these issues escalate. By recognizing patterns in student feedback, learning behavior, or course evaluations, institutions can tailor interventions, personalize support services, and revise instructional content to improve educational outcomes and the overall student experience.

In this paper, we propose a novel hybrid method called random tree with ordinal noise filtering and feature selection (RTONF). This method aims to build robust machine learning models by incorporating order-aware noise removal techniques with a systematic feature selection process, thereby enhancing model generalizability and prediction accuracy. RTONF leverages the interpretability and low variance of the random tree classifier, while the integration of ordinal noise filtering ensures that the natural order of class labels is preserved, leading to more meaningful and contextually accurate predictions in educational settings.

Fig. 1 presents the general architecture of the proposed RTONF method. As depicted, the process starts with the collection of raw survey data from students, which encompasses their feedback on course content, learning satisfaction, and instructor characteristics, using a 5-point Likert-type scale (poor $<$ fair $<$ good $<$ very good $<$ excellent). The architecture then proceeds to the data preprocessing stage, where ordinal noise detection is applied to filter out inconsistent or misleading survey responses, ensuring that the ordinal nature of the data is preserved. Next, a feature selection step (i.e., Pearson correlation) is performed to determine the most relevant factors that contribute to predicting student performance and satisfaction. Following data preprocessing, the random tree algorithm is employed as the core machine learning model, taking advantage of its robustness and interpretability. Subsequently, the model undergoes an evaluation phase using various performance metrics—including accuracy, recall, precision, f-measure, and confusion matrix—to assess its predictive quality. Finally, the trained model is utilized to make predictions on new student survey data, providing actionable visions that can guide institutional decision-making. This architecture integrates data collection, noise filtering, feature selection, machine learning, evaluation, and prediction into a unified framework, thereby facilitating practical educational data mining.

A. ORDINAL NOISE DETECTION

Student survey data often contains noisy or inconsistent values that require cleaning before analysis. The current noise detection methods are nominal, meaning that they do not consider the inherent order or ranking present in the data when identifying contradictory data instances. However, student surveys typically involve rating instructors and courses using scales such as {poor $<$ fair $<$ good $<$ very good $<$ excellent}, which are ordinal in nature and reflect a hierarchical order. This presents a challenge that is distinct from traditional (nominal) noise detection methods, which ignore the ordering among classes.

Unlike previous works, this study aims to detect noise in data on an ordinal scale. Specifically, we introduce a novel concept—ordinal noise detection—that meticulously adapts

TABLE 2. A dataset for an example scenario.

a	b	class	a	b	class	a	b	class	a	b	class	a	b	class	a	b	class	a	b	class	a	b	class
1	2	circle	2	12	square	3	14	square	5	3	circle	6	6	square	7	10	triangle	9	2	circle	11	8	triangle
1	12	square	2	13	square	4	1	circle	5	8	triangle	6	8	triangle	7	11	triangle	9	10	triangle	11	9	triangle
1	13	square	2	14	square	4	3	square	6	1	circle	6	10	triangle	8	1	circle	9	11	triangle	11	10	triangle
1	14	square	3	1	circle	4	13	square	6	4	circle	6	11	triangle	8	11	triangle	10	9	triangle			
2	2	circle	3	13	square	4	14	square	6	5	circle	7	8	triangle	9	1	circle	10	10	triangle			

the noise detection task to the ordinal setting. In this context, the noise detection task is carefully adapted to the ordinal scenario. The objective is to eliminate contradictory samples present in the dataset, thereby enhancing data quality. This innovative approach not only improves the quality of the dataset but also enhances the accuracy of the classifier more comprehensively. The general mechanism proceeds as follows:

- *Ordinal encoding*: The ordinal relationships among class labels are represented numerically, ranging from 1 to s, where s is the total number of classes. Labels are assigned increasing values from lowest to highest. For example, the categories {poor, fair, good, very good, excellent} are encoded as poor: 1, fair: 2, good: 3, very good: 4, and excellent: 5. This preserves the hierarchical relationship between categories and ensures that subsequent steps respect the order among the classes.
- *Neighborhood identification:* To identify potentially noisy samples, the pairwise distances between all samples are calculated using a suitable distance metric (e.g., Euclidean distance). For each sample, k-nearest neighbors are selected based on their distance. Each neighbor is assigned a proximity weight inversely proportional to its distance rank: the closest neighbor is given a weight of f k, the next closest $k-1$, and so on down to 1 for the furthest. This approach ensures that closer neighbors contribute more strongly to the noise detection process. The cumulative weight for k neighbors is thus the sum of the first k natural numbers, e.g., $1+2+3=6$ when $k=3$.
- *Noise detection:* The initial and basic condition for a noisy sample is having a different class label than its k closest neighbors. Using this rationale, for each neighbor from a different class, the absolute difference in encoded values is calculated. For example, if the sample is labeled ''excellent'' (5) and a neighbor is labeled ''good'' (3), the difference is 2. These label differences are weighted based on proximity, where closer neighbors have a greater impact. To standardize this difference, each weighted difference is divided by $s-1$, the maximum possible label difference. The sum of these normalized values across all differing neighbors gives the cumulative difference score. This score is further normalized by the cumulative weights for the k neighbors. For instance, if $k=3$, the cumulative weight is $1+2+3=6$. The final normalized score ranges between 0 and 1. If this score exceeds a predefined threshold (selected between 0 and 1), the sample is classified as noisy. To improve robustness, this process is repeated for multiple values of k, and majority voting across these results is used to make the final decision on whether a sample is considered noisy.

B. EXAMPLE FOR ORDINAL NOISE DETECTION

A toy dataset is given to illustrate the logic behind the proposed *ordinal noise detection* (OND) method in Table 2. The dataset has two features, a and b, and the class labels are circle, square, and triangle. The aim is to identify and remove noisy samples as depicted in Fig. 2. The dashed circles show the neighborhood area based on the number of nearest neighbors (k), while N_i, where i is from 1 to 9, denotes the ith nearest neighbor of a given sample X_1. As an example, the noise detection process using X_1 is represented in Table 3. In this process, four key parameters are considered: d, representing the difference between ordinal class labels; s, denoting the total number of classes; ω, which assigns a weight to each neighbor based on its distance rank; and k, the number of neighbors used in the evaluation. In the first step, all pairwise distances were calculated using the Manhattan distance. The k value was varied across values 1, 3, 5, and 9 to assess noise under different neighborhood sizes. For $k=1$, the closest neighbor of X_1 (located at (6, 6)) is N_1 (6, 5). X_1 has the class label ''square'', encoded as 1 (''Low''), and N_1 has the label ''circle'', encoded as 3 (''High''), resulting in a difference $d=3-1=2$. Weights (ω) were assigned in decreasing order from the nearest to the outermost neighbor. For example, when k was 5, N_1 (closest) received weight 5, and N_5 (farthest) received weight 1. Other neighbors' weights varied inversely with their distance from X_1. The total number of classes s is 3, so normalization is done using $s-1=2$. Each neighbor's contribution was calculated as $(d\times\omega)/(s-1)$. These contributions were then summed to get the total impact (Δ_k) for a given k. As an example, for $k=5$, the contributions from N_1 to N_5 summed to 11.5, and the total weight (W) was $5+4+3+2+1=15$. The final normalized score ($\mathbb{S}_k$) as $11.5/15=0.76$. A predefined threshold ($\theta=0.6$) was used to classify a sample as ''noise'' or ''not noise''. Since all normalized scores for X_1 across different k values exceeded the threshold, X_1 was classified as ''noise'' in every case. In the final step, a majority voting

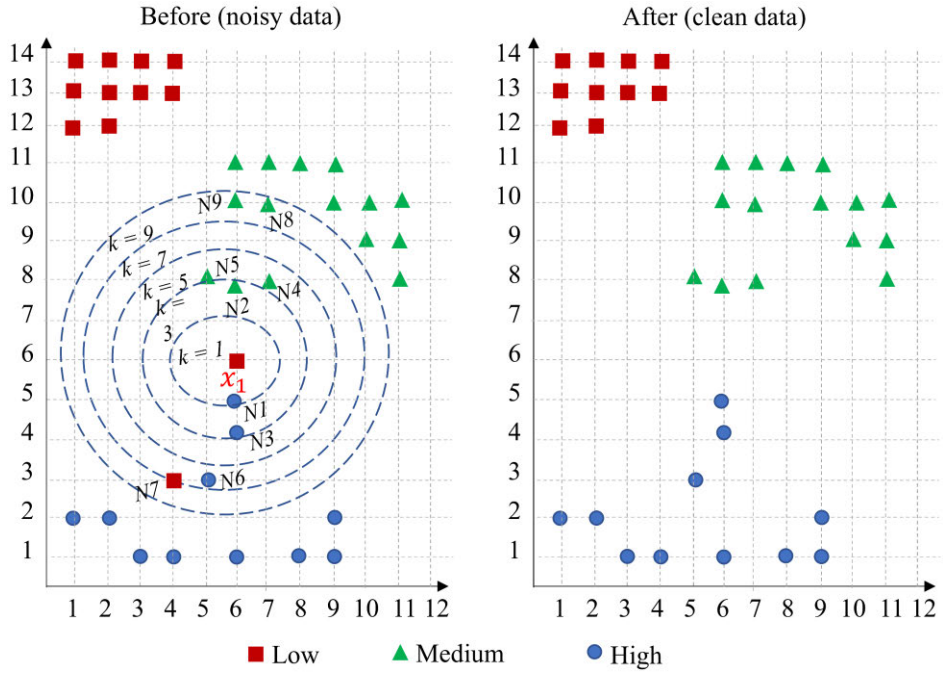


FIGURE 2. Illustration of the "*Ordinal Noise Detection*" process before and after noise removal for the example scenario.

TABLE 3. The noise identification process for a given sample.

k	N_1			N_2			N_3			N_4			N_5			N_6			N_7			N_8			N_9			Δ_k	W	$\mathbb{S}_k$	Decision
	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$	d	ω	$\frac{d\omega}{s-1}$				
1	2	1	1																									1.0	1	1.00	noise
3	2	3	3	1	2	1	2	1	1																			5.0	6	0.83	noise
5	2	5	5	1	4	2	2	3	3	1	2	1	1	1	0.5													11.5	15	0.76	noise
7	2	7	7	1	6	3	2	5	5	1	4	2	1	3	1.5	2	2	2	0	1	0							20.5	28	0.73	noise
9	2	9	9	1	8	4	2	7	7	1	6	3	1	5	2.5	2	4	4	0	3	0	1	2	1	1	1	0.5	31.0	45	0.68	noise

strategy was applied across all k values to make a consistent decision. Since all decisions agreed, X_1 was definitely labeled as noise. This same procedure was applied to all samples, and those classified as noise were removed to produce a cleaner dataset.

C. FEATURE SELECTION

Feature selection is a fundamental stage in the ML process to identify the most in-formative features for predictive modeling. In the current study, the Pearson correlation co-efficient was primarily employed to evaluate the relationship between individual features and the target variable. This technique is particularly suitable when dealing with continuous or interval-scaled data. The Pearson correlation coefficient quantifies the degree of dependence between two variables, ranging from -1 (perfect negative correlation) to $+1$ (perfect positive correlation). It is calculated as (1):

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{1}$$

where x_i and y_i denote the paired observations, $\bar{x}$ and $\bar{y}$ are the means of the respective variables, and n is the total number of observations. While our focus was on Pearson correlation, other feature selection methods can also be considered based on data characteristics and modeling goals. These include mutual information, recursive feature elimination, and feature importance scores derived from tree-based algorithms. Each approach has its own applicability, but for our analysis, Pearson correlation served as an interpretable measure to guide the selection of relevant features in the upcoming sections.

D. RANDOM TREE

The random tree classifier is a decision-tree-based model that introduces randomness into its construction process to enhance diversity and reduce overfitting. Unlike traditional decision trees, which deterministically consider all features and samples at each split, the random tree classifier injects stochasticity into feature selection and sample choice. This characteristic makes it particularly effective when employed as a base learner. Given a training dataset $D = \{(x_i, y_i)\}_{i=1}^{n}$, where $x_i \in \mathbb{R}^d$ denotes a feature vector and y_i is the cor-

responding label, the algorithm generates a set of candidate decision trees from the hypothesis space H. At each node of the tree, a random subset of features $F_m \subseteq F$ with $|F_m| = m \leq d$ is selected to determine the best split. This mechanism ensures that not all features influence every node, fostering model diversity.

Furthermore, the algorithm chooses an optimal split at each node using one of several criteria, depending on the task at hand. For classification, commonly used criteria include Gini impurity, which measures the degree of disorder or impurity within a node. For regression tasks, variance reduction is employed to quantify the spread of target values within a node. The tree construction process continues recursively, guided by stopping conditions such as a predefined maximum depth, a minimum sample size required to split a node, or a homogeneity of target values in a node. In classification, the final prediction at a leaf node is typically the majority class among the samples, whereas for regression, the output is the mean of the target values. The following equations define the key splitting criteria used:

- The Gini impurity for classification tasks: This metric quantifies the likelihood of misclassifying a randomly chosen sample from the node, as given in (2).

$$\text{Gini}(S) = 1 - \sum_{i=1}^{c} p_i^2 \tag{2}$$

  where S is the set of samples at a node, p_i is the proportion of samples of class i, and c is the number of classes.
- Variance reduction for regression tasks: This measure captures the spread of target values at a node, helping identify splits that lead to more homogeneous subgroups, as given in (3).

$$\text{Var}(S) = \frac{1}{|S|} \sum_{i \in S} (y_i - \bar{y})^2 \tag{3}$$

  where y_i is the target value for sample i and $\bar{y}$ is the mean of the target values in node S. This randomization approach not only helps prevent overfitting but also makes the model robust and suitable for a wide range of regression and classification tasks.

E. FORMAL DESCRIPTION

Let the dataset be represented as $\mathcal{D} = \{(X_1, y_1), (X_2, y_2), \ldots, (X_n, y_n)\}$ where n is the number of instances. Each ith row $X_i = (x_{i1}, x_{i2}, \ldots, x_{im}) \in \mathbb{R}^m$ refers to the feature vector of the ith instance, representing responses to m questionnaire items, e.g., $x_{ij} \in \{1, 2, 3, 4, 5\}$ representing the Likert-type scale (order) of preference of student i on the question j. It is noted that the Likert-type scale is obtained by mapping the response levels {poor, fair, good, very good, excellent} into pseudo-numbers $\{1, 2, 3, 4, 5\}$. Here, $y_i \in \{c_1, c_2, \ldots, c_s\}$ denotes the class label of instance i, where s is the number of classes. The overall procedure consists of three main stages, namely ordinal noise filtering, feature selection, and classification.

- Ordinal noise filtering aims to identify and remove potentially mislabeled instances by exploiting ordinal inconsistencies within local neighborhoods. The procedure proceeds as follows:
  i. *Neighborhood identification*: Let $\delta: \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}_{\geq 0}$ be a distance function that measures dissimilarity between instances (e.g. Euclidean, Manhattan, or Cosine distance). For each instance $X_i \in \mathbb{R}^m$, the distance to all other instances $X_j \in D \setminus \{X_i\}$ is computed as (4).

$$\delta_{ij} = \delta\left(X_i, X_j\right) \tag{4}$$

  Using these pairwise distances, the k-nearest neighbors of X_i is identified as $\mathcal{N}_k(X_i) = \{X_{i1}, X_{i2}, \ldots, X_{ik}\}$ such that the following (5):

$$\delta(X_i, X_{i1}) \leq \delta(X_i, X_{i2}) \leq \ldots \leq \delta(X_i, X_{ik}) \tag{5}$$

  where X_{ij} refers to the jth closest neighbor to X_i based on the selected distance function δ. Each neighbor $X_{ij} \in \mathcal{N}_k(X_i)$ is assigned a rank-based weight ω_j to emphasize closer neighbors. The weights are defined as (6):

$$\omega_j = k - j + 1, \quad \text{for } j = 1, 2, \ldots, k \tag{6}$$

  The weighting schema gives the highest weight to the closest neighbor ($j = 1$) and the lowest weight to the farthest neighbor ($k = 1$).
  ii. *Noise scoring and label discrepancy*: Let y_i and y_{ij} denote the numerically encoded class labels of X_i and its jth nearest neighbor X_{ij}, respectively. For neighbors with $y_i \neq y_i$, compute the weighted normalized ordinal discrepancy as (7)

$$d_{ij} = \frac{\omega_j \cdot |y_i - y_{ij}|}{s - 1} \tag{7}$$

  Summing all the values from the list of neighbors in $\mathcal{N}_k(X_i)$ gives the total discrepancy score for X_i as (8):

$$\Delta_k(X_i) = \sum_{j=1}^{k} d_{ij} \tag{8}$$

  Normalize this score by the total weight as (9):

$$W_k = \sum_{j=1}^{k} \omega_j = \frac{k(k+1)}{2},$$
$$\mathbb{S}_k(X_i) = \frac{\Delta_k(X_i)}{W_k} \tag{9}$$

  where $\mathbb{S}_k(X_i) \in [0, 1]$. Here $\mathbb{S}_k(X_i)$ represents the ordinal inconsistency score of X_i for neighborhood size k.
  iii. *Noise filtering via thresholding and voting*: Let $\theta \in [0, 1]$ be a predefined threshold. An instance is flagged as noisy for a given k based on (10):

$$\eta_k(X_i) = \begin{cases} 1 & \mathbb{S}_k(X_i) \geq 0 \\ 0 & otherwise \end{cases} \tag{10}$$

To improve robustness, this evaluation is repeated for multiple neighborhood sizes $\mathcal{K} = \{k_1, k_2, \ldots, k_r\}$, typically odd integers. The final noise label is determined by majority voting based on (11):

$$\eta(X_i) = \begin{cases} 1 & \sum_{k \in \mathcal{K}} \eta_k(X_i) > \frac{|\mathcal{K}|}{2} \\ 0 & otherwise \end{cases} \quad (11)$$

Instances for which $\eta(X_i) = 1$ are classified as noisy and are removed from the training set. $\mathcal{D}' = \{(X_1, y_1), (X_2, y_2), \ldots, (X_{n'}, y_{n'})\} \subset \mathcal{D}$ denotes the cleaned dataset obtained after the removal of noisy instances, where $n' \leq n$.

- *Feature selection:* To reduce dimensionality and enhance discriminative capacity, a feature selection function $\mathcal{F}$ is applied as (12):

$$\mathcal{F} : \mathbb{R}^m \to \mathbb{R}^{m'}, \quad \text{where } m' < m \quad (12)$$

This transformation selects a subset $\mathcal{S} \subset \{1, 2, \ldots, m\}$ of the original features, such that (13):

$$X_i' = \mathcal{F}(X_i)\left(x_{ij_1}, x_{ij_2}, \ldots, x_{ij_{m'}}\right), \quad \text{for } j_k \in \mathcal{S} \quad (13)$$

The function $\mathcal{F}$ ranks features using a relevance criterion $\psi\left(x_j, Y\right)$, such as the Pearson correlation, information gain, or mutual information. The top m' features with the highest values of ψ are selected to form the new reduced feature set $X' \in \mathbb{R}^{n' \times m'}$.

- *Classification*: Let $\mathcal{D}^T = \{(X_1', y_1), (X_2', y_2), \ldots, (X_{n'}, y_{n'})\}$ denotes the training set after noise removal and feature selection, where $X_i' \in \mathbb{R}^{m'}$ is the reduced feature vector for the ith instance and $y_i \in \{1, 2, \ldots, s\}$ is the corresponding ordinal class label. A decision tree-based classifier, denoted by the mapping function h:$\mathbb{R}^{m'} \to \{1, 2, \ldots, s\}$ is trained on $\mathcal{D}^T$ using the Random Tree algorithm. The learned model h maps new instances to predicted class labels. Let $\mathcal{T} = \left\{\left(X_1^{test}, y_1^{true}\right), \ldots, \left(X_q^{test}, y_q^{true}\right)\right\}$ be the test set, disjoint from $\mathcal{D}^T$. The predicted labels are given by (14):

$$y_i^{pred} = h\left(X_i^{test}\right), \quad \text{for } i = 1, \ldots, q \quad (14)$$

F. ALGORITHM

The proposed RTONF method is a robust, hybrid machine learning framework that integrates ordinal noise filtering, feature selection, and classification to effectively analyze student survey data while ensuring model interpretability. It operates on an ordinal dataset $\mathcal{D} = \{(X_1, y_1), (X_2, y_2), \ldots, (X_n, y_n)\}$, where each X_i represents a vector of student responses on a Likert-type scale (mapped numerically from 1 to 5), and y_i denotes the ordinal class label, such as attendance levels. The method proceeds in three main stages: (i) ordinal noise filtering, (ii) feature selection, and (iii) classification.

i. In the ordinal noise filtering stage, each instance is evaluated by measuring ordinal inconsistencies within its local neighborhood. Using a distance function (e.g., Euclidean), the k-nearest neighbors $\mathcal{N}_k(X_i)$ are identified and ranked by distance. A rank-based weighting scheme ($\omega_j = k - j + 1$) emphasizes closer neighbors. For neighbors with differing class labels, a weighted normalized ordinal discrepancy d_{ij} is computed to capture how different and distant the neighbor's label is. Summing across all neighbors yields the total discrepancy $\Delta_k(X_i)$, which is then normalized by the total weight ω_k to obtain an ordinal inconsistency score $\mathbb{S}_k(X_i) \in [0, 1]$. If $\mathbb{S}_k(X_i)$ exceeds a predefined threshold θ, the instance is flagged as noisy for that neighborhood size. To ensure reliability, this evaluation is repeated across multiple neighborhood sizes $\mathcal{K} = \{k_1, k_2, \ldots, k_r\}$, and majority voting determines the final noise label. Instances flagged as noisy are removed from the dataset, yielding a cleaned dataset D'.

ii. Next, in the feature selection stage, a feature selection function F (e.g., Pearson correlation) ranks features by their relevance to the class labels. The top m' features are selected to form a reduced feature set, to improve discriminative capacity, and to reduce dimensionality.

iii. Finally, in the classification stage, a random tree classifier h is trained on the cleaned and reduced dataset D^T. This classifier constructs decision trees by selecting splits that maximize purity, ensuring that ordinal structure is respected. For testing, each new instance in the test set $\mathcal{T}$ is fed into the trained model h to predict its class label.

Algorithm 1 summarizes this entire process; it iterates over each training instance, computes weighted ordinal discrepancy scores across multiple neighborhood sizes, flags noisy instances, performs feature selection, and trains a random tree classifier. Finally, it predicts the class labels of test instances. By integrating ordinal noise detection and feature selection before classification, RTONF improves efficiency against label noise while maintaining interpretability and high predictive performance, making it well-suited for analyzing ordinal survey data such as student satisfaction and attendance.

The complexity of the algorithm is $O(n \log n) + T(n)$, where n is the number of instances and $T(n)$ indicates the time needed for the execution of the ordinal noise identification process. This complexity is consistent with the random tree algorithm, reflecting the efficiency of the proposed method.

IV. EXPERIMENTS

The proposed RTONF approach was developed using the C# programming language and integrated with the WEKA framework [52] to facilitate model training and validation. Experimental procedures were conducted on a conventional desktop system equipped with an Intel® Core™ i7 processor (1.90 GHz) and 8 GB of memory. For predicting outcomes related to the student survey dataset, the method adopted the random tree algorithm as its foundational classifier.

Algorithm 1 Random Tree With Ordinal Noise Filtering and Feature Selection (RTONF)

Inputs:
D: The ordinal dataset with n instances such that $D = \{(X_1, y_1), (X_2, y_2), \ldots, (X_n, y_n)\}$
Y: Ordinal class labels $y \in \{c_1, c_2, \ldots, c_s\}$ with an order $c_1 < c_2 < \ldots < c_s$
m': Number of features to be selected
θ : The minimum value (thresholds) for an instance to be defined as noise
$\mathcal{K}$: The neighborhood sizes to be applied in each iteration that $\mathcal{K} = \{k_1, k_2, \ldots, k_r\}$
$\mathcal{F}$: Feature selection method, i.e., Pearson Correlation
$\mathcal{T}$: New instances to be predicted
Outputs:
$\hat{Y}$: Class labels predicted for the inputs in $\mathcal{T}$

// Step 1 – Ordinal Noise Filtering
foreach(x, y) **in** D
  $W = 0$
  noise = 0
  for $k = 1$ **to** r **with step 2 do**
  $\mathcal{N}_k(x) = \text{NNSearch}(x, k)$
  Sort($\mathcal{N}_k(x)$)
  $\Delta_k = 0$
  for $i = 0$ **to** *k-1*
    if $\mathcal{N}_k(x)$[i]. $y \neq y$
      $\Delta_k = \Delta_k + (k - i)^* |\mathcal{N}_k(x)[i].y - y|/(s - 1)$
    endif
  end for
  $W = W + k$
  $\mathbb{S}_k = \Delta_k / W$
  if $\mathbb{S}_k >= \theta$
    noise = noise + 1
  end if
 end for
 if noise < length($\mathcal{K}$) / 2
    $\mathcal{D}'$.Add(x, y)
end foreach
// Step 2 – Feature Selection
$\mathcal{D}^{\mathcal{T}} = \mathcal{F}(\mathcal{D}', m')$
// Step 3 - Training
$h = \text{RandomTree}(\mathcal{D}^{\mathcal{T}})$
// Step 4 - Testing
foreach x **in** T
  $y = h(x)$
  $\hat{Y} = \hat{Y} \cup y$
end foreach
End Algorithm

To assess the performance of RTONF, a 10-fold cross-validation strategy was adopted, a standard technique for estimating how well a model generalizes to independent data. In this method, the dataset is randomly divided into ten equal parts. For each iteration, nine subsets are used for model training, and the remaining one is reserved for testing. The process is repeated ten times, ensuring that each subset serves as the test set exactly once. The final evaluation is obtained by averaging the results across all folds, providing a robust and unbiased performance estimate.

In addition to cross-validation, a comprehensive experimental analysis was conducted to assess the predictive strength of the proposed model. We employed several evaluation metrics—namely accuracy, recall, precision, and F-measure—to compare RTONF with alternative classification methods. These metrics rely on fundamental classification outcomes, including true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) [53]. Their formulations are outlined in (15) to (18), respectively:

- Accuracy is defined as the proportion of correctly predicted instances (both positive and negative) to the total number of instances evaluated. It serves as a general indicator of classification success:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{15}$$

- Precision measures the proportion of correctly predicted positive cases out of all instances labeled as positive by the model. This metric is particularly useful in scenarios with class imbalance, where minimizing false positives is crucial:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{16}$$

- Recall captures the model's ability to correctly identify actual positive instances:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{17}$$

- F-measure (or F1-score) is the harmonic mean of precision and recall, offering a balanced evaluation that considers both false negatives and false positives:

$$\text{Fmeasure} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

A. DATASET DESCRIPTION

In the scope of this study, we selected the Turkiye student evaluation dataset (TSED) from the University of California, Irvine (UCI) ML repository [54] as a real-world dataset. This dataset measures various aspects of students' general satisfaction with courses and instructors at Gazi University in Ankara (Turkey). Table 4 presents a summary of the dataset. The data consists of 5,820 instances and 33 attributes. Among these, 28 attributes correspond to course-specific questions, while the remaining five attributes capture information such as the instructor identifier, class, the number of times a student has repeated the course, attendance level, and perceived difficulty of the course. Questions 1 to 7 focus on course content, Questions 8 to 12 assess student satisfaction with learning activities, and Questions 13 to 28 evaluate the instructor's characteristics. The course-specific questions are rated on a Likert-type scale with response categories — {poor, fair, good, very good, excellent} — encoded as ordinal values {1, 2, 3, 4, 5}.

Since the dataset contains no missing values across any of the attributes, data imputation is not required. Some students have enrolled in the particular course two or three times, so the response to the number of repeats is an integer value between 1 and 3. The dataset has an imbalance in the number of repeats, i.e., '1' is a fairly dominant value rather than the values of '2' and '3'. It is important to consider this issue during building a machine learning model since the classification performance can be affected by the imbalance problem if present in the dataset. To address this challenge, the attribute values are considered as ''no'' if the student is taking the course for the first time, ''yes''

TABLE 4. Attribute information of the TSED dataset.

ID	Name	Description	Values
1	instr	Instructor's identifier.	{1,2,3}
2	class	Course code.	{1-13}
3	repeat	Number of times the student is taking this course.	{1,2,3}
4	attendance	Code of the level of attendance.	{0,1,2,3,4}
5	difficulty	Level of difficulty of the course as perceived by the student	{1,2,3,4,5}
6	Q1	The semester course content, teaching method and evaluation system were provided at the start.	{1,2,3,4,5}
7	Q2	The course aims and objectives were clearly stated at the beginning of the period.	{1,2,3,4,5}
8	Q3	The course was worth the amount of credit assigned to it.	{1,2,3,4,5}
9	Q4	The course was taught according to the syllabus announced on the first day of class.	{1,2,3,4,5}
10	Q5	The class discussions, homework assignments, applications and studies were satisfactory.	{1,2,3,4,5}
11	Q6	The textbook and other course resources were sufficient and up to date.	{1,2,3,4,5}
12	Q7	The course allowed fieldwork, applications, laboratory, discussion and other studies.	{1,2,3,4,5}
13	Q8	The quizzes, assignments, projects and exams contributed to helping the learning.	{1,2,3,4,5}
14	Q9	I greatly enjoyed the class and was eager to actively participate during the lectures.	{1,2,3,4,5}
15	Q10	My initial expectations about the course were met at the end of the period or year.	{1,2,3,4,5}
16	Q11	The course was relevant and beneficial to my professional development.	{1,2,3,4,5}
17	Q12	The course helped me look at life and the world with a new perspective.	{1,2,3,4,5}
18	Q13	The Instructor's knowledge was relevant and up to date.	{1,2,3,4,5}
19	Q14	The Instructor came prepared for classes.	{1,2,3,4,5}
20	Q15	The Instructor taught in accordance with the announced lesson plan.	{1,2,3,4,5}
21	Q16	The Instructor was committed to the course and was understandable.	{1,2,3,4,5}
22	Q17	The Instructor arrived on time for classes.	{1,2,3,4,5}
23	Q18	The Instructor has a smooth and easy-to-follow delivery/speech.	{1,2,3,4,5}
24	Q19	The Instructor made effective use of class hours.	{1,2,3,4,5}
25	Q20	The Instructor explained the course and was eager to be helpful to students.	{1,2,3,4,5}
26	Q21	The Instructor demonstrated a positive approach to students.	{1,2,3,4,5}
27	Q22	The Instructor was open and respectful of the views of students about the course.	{1,2,3,4,5}
28	Q23	The Instructor encouraged participation in the course.	{1,2,3,4,5}
29	Q24	The Instructor gave relevant homework assignments/projects, and helped/guided students.	{1,2,3,4,5}
30	Q25	The Instructor responded to questions about the course inside and outside of the course.	{1,2,3,4,5}
31	Q26	The Instructor's evaluation system (midterm and final questions, projects, assignments, etc.), effectively measured the course objectives.	{1,2,3,4,5}
32	Q27	The Instructor provided solutions to exams and discussed them with students.	{1,2,3,4,5}
33	Q28	The Instructor treated all students in a right and objective manner.	{1,2,3,4,5}

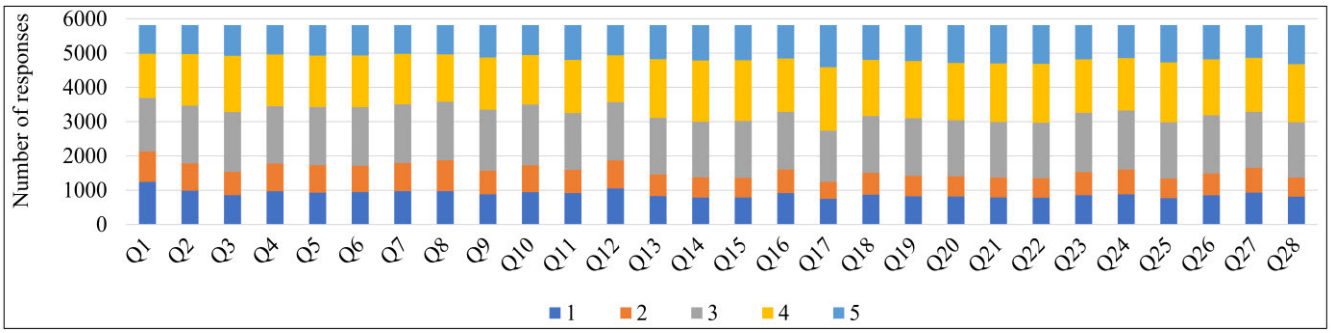


FIGURE 3. Number of responses for the course-specific questions Q1 to Q28.

otherwise [55], [56], [57], [58], [59]. Furthermore, similar to the previous studies [60], the values of attendance level to the course are taken as {0, 1, 2, 3, 4}, whereas values less than 2 are considered to be low, values equal to or greater than 2 are noted as high. Moreover, to be comparable with previous studies [60], [61], we categorized difficulty scores of {1, 2} as below-average and {3, 4, 5} as above-average.

Fig. 3 presents a stacked bar chart illustrating the distribution of student feedback scores, ranging from 1 (strongly disagree) to 5 (strongly agree), across 28 survey questions (Q1 to Q28). Each bar represents the total number of responses received for a specific question, segmented by score. The majority of responses are concentrated in the higher score categories (3, 4, and 5), indicating that students generally responded positively to most items. Notably, the proportion of ratings at levels 4 and 5 is consistently high across all questions, suggesting a strong level of satisfaction with various aspects of the course. The relatively small portion of scores at levels 1 and 2 also reinforces this trend, implying limited dissatisfaction. Overall, the chart provides a comprehensive overview of how students evaluated different

course components, showcasing areas of strength and consistency in student perceptions.

V. RESULTS

A. ACCURACY ANALYSIS

A comparative analysis of prediction accuracies between the baseline random tree (RT) classifier and the proposed RTONF method across 32 distinct target class attributes of the TSED dataset is presented in Table 5. The attributes include general class variables such as instructor, repeat, attendance, and difficulty, as well as individual survey items labeled Q1 through Q28. The best-performing results are marked with a • symbol. The results clearly proved that RTONF, with an average accuracy of 84.61%, consistently outperformed the baseline RT model, with an average accuracy of 80.42%, across all target class attributes, indicating the superiority of RTONF with a 5.23% improvement. For example, when predicting the ''instr'' attribute, RTONF achieved 98.26%, significantly surpassing the baseline accuracy of 90.33%. For the ''repeat'' attribute, a notable improvement from 80.45% to 84.35% was observed. The ''attendance'' attribute also saw a substantial gain, improving from 65.52% to 71.46%, showcasing RTONF's success even on more challenging attributes. Across the 28 survey-related questions (Q1–Q28), RTONF improved accuracy by margins ranging from approximately 3% to 8% over the baseline model. The largest relative improvement occurred in Q1, where the accuracy increased from 75.52% to 83.06%, a 9.98% absolute gain. These consistent advances point out the benefits of incorporating ordinal noise filtering and feature selection into the classification pipeline. In particular, the results suggest that ordinal-aware noise removal boosts the ability of the model to generalize, especially in scenarios involving subjective and highly variable student feedback. Mostly, the fact that RTONF achieved the highest accuracy for all 32 attributes (as indicated by the • symbol) underscores its adaptability across diverse prediction targets in educational data mining.

The performance gain (%) achieved by the proposed RTONF method over the baseline RT classifier across 32 different target class attributes is illustrated in Fig. 4. Performance gain was computed as the relative improvement in classification accuracy, measured as the increase in accuracy achieved by RTONF compared to RT, expressed as a percentage relative to the RT accuracy. As shown in the chart, the RTONF method consistently outperformed the baseline across all target variables, with improvements ranging from 2.52% to 9.98%. The most substantial gain was observed when Q1 was selected as the target class, showing a 9.98% improvement.

Other notable gains include ''instr'' (8.78%), Q2 (6.51%), and Q27 (6.28%), illustrating RTONF's robustness across both student evaluation questions and key performance indicators such as instructor quality and course difficulty. This consistent upward trend reinforces RTONF's ability to mitigate noise while increasing predictive ability. The stable performance gains across all class attributes, with an average 5.23% improvement, suggest that the method is broadly applicable within educational data mining tasks, without overfitting to specific features.

The statistical tests were done to establish the significance and effectiveness of the proposed method. The results of the statistical analysis demonstrated significant differences between the compared groups, as determined by both the Wilcoxon test and the Mann-Whitney U test, evaluated at a significance level of $\alpha = 0.05$. The Wilcoxon test yielded a p-value of approximately 7.95×10^{-7}, and the Mann-Whitney U test produced a p-value of approximately 5.15×10^{-6}. Since both p-values are well below the 0.05 threshold, the null hypotheses (H_0), which assume no difference between the groups, are rejected for both tests. These findings indicate strong statistical evidence of a difference in the distributions of the compared samples.

B. PRECISION, RECALL, AND F-MEASURE ANALYSIS

A detailed comparison of the baseline RT and the proposed RTONF across 32 different target class attributes is presented in Table 6 in terms of other core classification metrics. Across all attributes, the proposed method steadily outperformed the baseline with an average precision of 0.8467, recall of 0.8461, and F-measure of 0.8464. This consistency suggests that the proposed noise filtering method successfully enriches the classifier's ability to both accurately identify positive instances (precision) and detect a larger portion of true positives (recall), leading to a better balance between the two (F-measure). For example, the improvement was particularly striking for the ''instr'' attribute, in which precision increased from 0.9028 to 0.9831, recall from 0.9033 to 0.9826, and F-measure from 0.9030 to 0.9829. These near-perfect values suggest that RTONF can handle structured, categorical attributes with minimal ambiguity very well. The ''repeat'' and ''attendance'' variables also indicated significant gains, with F-measure improvements of 6.98 and 9.30 percentage points, respectively. This determines the method's strength in predicting student behavioral factors, which are often related to ordinal noise patterns. For the ''difficulty'' attribute, RTONF showed moderate but consistent improvements across all metrics, again emphasizing its capability to refine predictions in the presence of ambiguous or subjective data.

Improvements were also observed across the 28 Likert-scale questionnaire attributes (Q1–Q28). For example, most metrics improved by approximately 2–9%, which is noteworthy given the ordinal nature and potential noisiness of such survey data. As another example, attribute Q1 saw its F-measure rise from 0.7550 to 0.8315—an improvement of 10.13%—emphasizing the method's potential for handling ordinal survey data with subjective bias. Besides, attributes such as Q14, Q21, and Q7 showed some of the highest performance gains, affirming RTONF's utility in real-world education survey contexts where class boundaries are often fuzzy. Importantly, RTONF does not dispropor-

TABLE 5. Accuracy comparison between the baseline RT classifier and the proposed RTONF method across various target class attributes.

ID	Target Class Attribute	RT	RTONF	ID	Target Class Attribute	RT	RTONF
1	instr	90.33	98.26 ●	17	Q13	82.49	87.47 ●
2	repeat	80.45	84.35 ●	18	Q14	83.97	88.97 ●
3	attendance	65.52	71.46 ●	19	Q15	83.90	88.61 ●
4	difficulty	69.05	71.67 ●	20	Q16	81.17	85.69 ●
5	Q1	75.52	83.06 ●	21	Q17	82.71	85.53 ●
6	Q2	78.61	83.73 ●	22	Q18	81.89	85.57 ●
7	Q3	77.97	81.84 ●	23	Q19	82.30	85.77 ●
8	Q4	79.35	82.16 ●	24	Q20	83.25	87.04 ●
9	Q5	80.27	83.11 ●	25	Q21	84.60	89.04 ●
10	Q6	79.45	83.78 ●	26	Q22	85.55	88.04 ●
11	Q7	79.86	84.76 ●	27	Q23	82.92	86.86 ●
12	Q8	80.27	83.38 ●	28	Q24	81.79	85.89 ●
13	Q9	77.78	82.22 ●	29	Q25	82.49	85.62 ●
14	Q10	81.32	83.37 ●	30	Q26	81.55	85.48 ●
15	Q11	78.08	82.13 ●	31	Q27	79.73	84.74 ●
16	Q12	77.68	82.20 ●	32	Q28	81.68	85.69 ●



FIGURE 4. Performance gain (%) achieved by the proposed RTONF method compared to the baseline random tree classifier across different target class attributes.

tionately optimize one metric at the expense of another. Instead, it maintains a balanced improvement across precision, recall, and F-measure, which is crucial in avoiding overfitting or skewed predictions. This balance confirms that the method not only makes more confident predictions but also generalizes better across different classes and data types. By efficiently detecting and mitigating ordinal noise, the RTONF method improves the quality of the training data, leading to great predictive power.

C. METRIC-WISE PERFORMANCE COMPARISON

The comparative bar chart clearly illustrates the steady performance boosts achieved by the proposed RTONF approach across four key evaluation metrics: accuracy, precision, recall, and F-measure in Fig. 5. Notably, the average accuracy increased from 80.42% (RT) to 84.61% (RTONF) with a 5.23% improvement, underlining the model's higher ability to correctly classify instances across diverse target attributes. This gain reinforces the overall reliability of the proposed approach. In terms of precision, the RTONF model reached an average of 84.67%, compared to 80.33% by the baseline RT with a 5.43% improvement. This indicates that the proposed method is more effective at reducing false positives, thereby improving confidence in its positive predictions. The average recall was equally noticeable, rising from 80.42% (RT) to 84.61% (RTONF) with a 5.23% improvement. This metric reflects the model's capacity to capture a higher proportion of actual positive instances, suggesting that RTONF helps uncover subtle patterns otherwise missed by the baseline. Lastly, the F-measure, which balances recall and precision, also showed a consistent increase from 80.38% to 84.64% with a 5.33% improvement. This advance confirms that the proposed method does not merely favor one performance aspect over another, but contributes to overall model

TABLE 6. Precision, Recall, and F-measure comparison between the random tree (baseline) and the proposed RTONF method across different target class attributes.

ID	Target Class Attribute	Precision		Recall		F-measure	
		RT	RTONF	RT	RTONF	RT	RTONF
1	instr	0.9028	0.9831 •	0.9033	0.9826 •	0.9030	0.9829 •
2	repeat	0.7732	0.8435 •	0.8045	0.8435 •	0.7885	0.8435 •
3	attendance	0.6553	0.7177 •	0.6552	0.7146 •	0.6552	0.7161 •
4	difficulty	0.6874	0.7113 •	0.6905	0.7167 •	0.6890	0.7140 •
5	Q1	0.7548	0.8325 •	0.7552	0.8306 •	0.7550	0.8315 •
6	Q2	0.7859	0.8411 •	0.7861	0.8373 •	0.7860	0.8392 •
7	Q3	0.7808	0.8180 •	0.7797	0.8184 •	0.7802	0.8182 •
8	Q4	0.7931	0.8209 •	0.7935	0.8216 •	0.7933	0.8213 •
9	Q5	0.8029	0.8312 •	0.8027	0.8311 •	0.8028	0.8312 •
10	Q6	0.7949	0.8394 •	0.7945	0.8378 •	0.7947	0.8386 •
11	Q7	0.7986	0.8482 •	0.7986	0.8476 •	0.7986	0.8479 •
12	Q8	0.8028	0.8362 •	0.8027	0.8338 •	0.8028	0.8350 •
13	Q9	0.7779	0.8252 •	0.7778	0.8222 •	0.7779	0.8237 •
14	Q10	0.8132	0.8330 •	0.8132	0.8337 •	0.8132	0.8333 •
15	Q11	0.7803	0.8241 •	0.7808	0.8213 •	0.7805	0.8227 •
16	Q12	0.7769	0.8213 •	0.7768	0.8220 •	0.7769	0.8216 •
17	Q13	0.8257	0.8754 •	0.8249	0.8747 •	0.8253	0.8751 •
18	Q14	0.8401	0.8890 •	0.8397	0.8897 •	0.8399	0.8894 •
19	Q15	0.8399	0.8863 •	0.8390	0.8861 •	0.8394	0.8862 •
20	Q16	0.8123	0.8571 •	0.8117	0.8569 •	0.8120	0.8570 •
21	Q17	0.8273	0.8585 •	0.8271	0.8553 •	0.8272	0.8569 •
22	Q18	0.8196	0.8545 •	0.8189	0.8557 •	0.8192	0.8551 •
23	Q19	0.8234	0.8580 •	0.8230	0.8577 •	0.8232	0.8579 •
24	Q20	0.8328	0.8714 •	0.8325	0.8704 •	0.8326	0.8709 •
25	Q21	0.8463	0.8904 •	0.8460	0.8904 •	0.8462	0.8904 •
26	Q22	0.8553	0.8812 •	0.8555	0.8804 •	0.8554	0.8808 •
27	Q23	0.8287	0.8682 •	0.8292	0.8686 •	0.8290	0.8684 •
28	Q24	0.8179	0.8600 •	0.8179	0.8589 •	0.8179	0.8595 •
29	Q25	0.8258	0.8579 •	0.8249	0.8562 •	0.8253	0.8571 •
30	Q26	0.8151	0.8547 •	0.8155	0.8548 •	0.8153	0.8547 •
31	Q27	0.7975	0.8481 •	0.7973	0.8474 •	0.7974	0.8478 •
32	Q28	0.8175	0.8569 •	0.8168	0.8569 •	0.8172	0.8569 •

robustness. Together, these average results demonstrate that the ordinal noise filtering strategy systematically develops the classification quality across all considered evaluation metrics, validating its general applicability in the context of educational data classification.

D. CONFUSION MATRIX ANALYSIS

While previous subsections focused on general performance metrics such as accuracy, precision, recall, and F-measure, confusion matrix analysis provides a more detailed view of the model's classification behavior by showing the exact distribution of correct and incorrect predictions across each rating class. This deeper layer of evaluation contributes to identifying not only how well the model performs in general but also where it tends to confuse specific classes, principally important in ordinal data such as Likert-scale feedback.

The confusion matrices for each of the 28 course-specific survey questions (Q1–Q28) are shown in Fig. 6 to capture how frequently the predicted labels match the actual responses in the five-point Likert scale (1 to 5). Each matrix illustrates the distribution of classification outcomes, with darker shades indicating higher frequencies. The diagonal elements represent correct predictions, while off-diagonal cells indicate misclassifications. A clear observation across

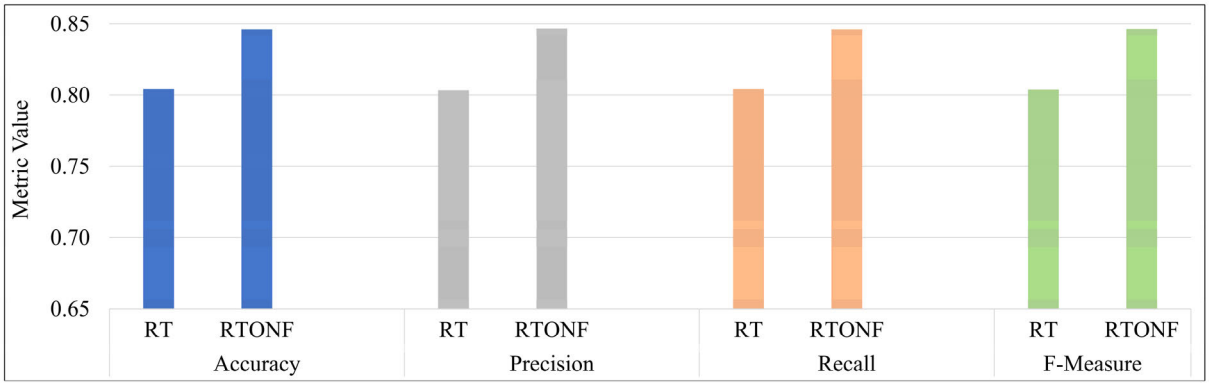


FIGURE 5. Comparison of average accuracy, precision, recall, and F-measure between the baseline random tree and the proposed RTONF approach.

most matrices is the dominance of diagonal values, particularly for classes 3, 4, and 5, suggesting that the model tends to predict middle-to-high satisfaction ratings with relatively high accuracy. Misclassifications are generally concentrated between neighboring scores (e.g., a true label of 4 predicted as 3 or 5), which is expected in ordinal classification problems due to the semantic similarity between adjacent labels.

Among all the questions, Q21 stands out with particularly strong predictive performance. Its confusion matrix reveals that the majority of instances fall along the diagonal in the higher rating categories. The number of correct predictions for scores 3, 4, and 5 is noticeably high, with minimal confusion among the lower scores. This suggests that responses to Q21 are more consistently patterned and easier for the model to learn. In the main, the confusion matrix analysis proves the reliability of the proposed method, which handles the ordinal noise removal task.

E. FEATURE IMPORTANCE

In this study, the Pearson's correlation coefficient was utilized to evaluate and rank the importance of features. This analysis allowed us to reduce the number of factors under consideration, which in turn improved both training speed and prediction accuracy. To address the possibility of nonlinear relationships in the data, we also tested other feature selection methods, including Information Gain, Gain Ratio, and ReliefF. These methods are designed to capture nonlinear dependencies between the features and the target variable. The resulting accuracies were 84.50%, 84.55%, and 83.67%, respectively, compared to 84.61% with Pearson correlation. Since Pearson correlation slightly outperformed the others in our case, this suggests that the most relevant relationships in the dataset are largely linear. For this reason, Pearson correlation was decided to be retained as the main feature selection method, as it provides both simplicity and efficiency without reducing model performance.

Table 7 shows the selected features for each target class attribute after applying the Pearson' correlation. The results revealed meaningful interrelationships among the top-ranked variables. For example, Q13 (Instructor's knowledge was relevant and up to date) was found to be highly correlated with Q14 (Instructor came prepared for classes), suggesting a strong link between the instructor's expertise and their preparedness. Additionally, Q14 demonstrated significant correlations with Q16 (Instructor was committed to the course and was understandable), Q15 (Instructor taught in accordance with the announced lesson plan), and Q17 (Instructor arrived on time for classes). These relationships highlight clusters of instructor-related variables that collectively influence students' perceptions of teaching quality. The instructor-related variables (Q13–Q28) consistently emerged as selected features across different target class attributes, underscoring their central role in shaping student satisfaction. Selected features for each target class attribute, emphasizing the overlap of key factors and reinforcing the importance of both instructor efficacy and course engagement in predicting student satisfaction outcomes.

As represented in Table 7, attributes such as "repeat", "attendance", and "difficulty" are often selected each other as important features, reflecting their interconnectedness in influencing course engagement and overall learning experience. By controlling for Q5, high predictive performance was achieved through the three important variables (Q4, Q6, and Q7), further underscoring their collective importance in determining student satisfaction. Remarkably, Q5 emerged as a key driver of student satisfaction, with the evaluation of the class homework assignments, discussions, studies, and applications.

F. PARAMETER ANALYSIS

To evaluate the sensitivity of the RTONF method to the threshold parameter θ, we conducted an empirical analysis by systematically varying θ from 0.1 to 0.8 and recording the corresponding average accuracy values. The results are presented in Fig. 7. As the threshold increased, a gradual improvement in average accuracy was observed, reaching its peak at $\theta = 0.6$. The results indicate that the RTONF method exhibits moderate sensitivity to θ, where small increases in

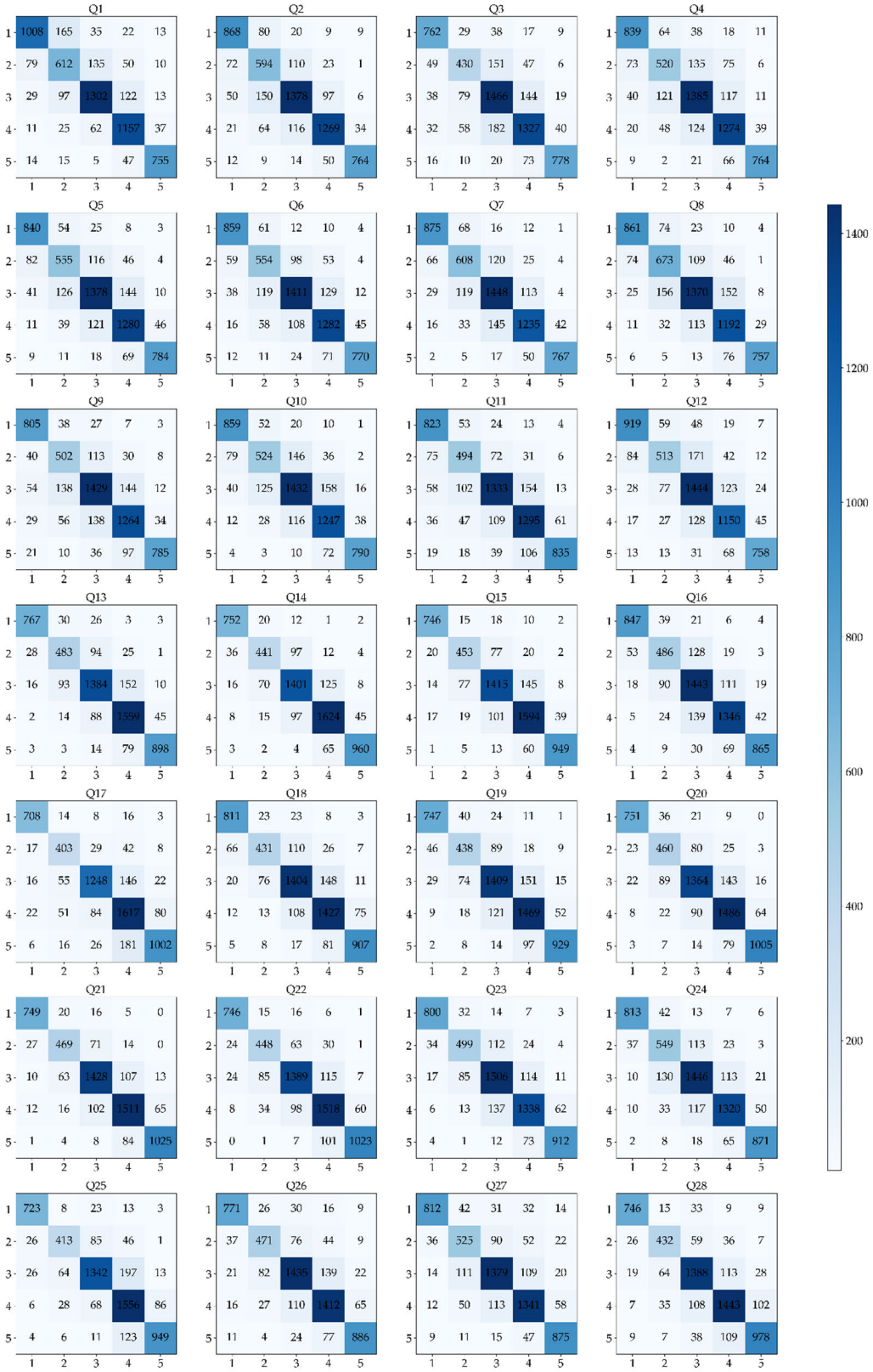


FIGURE 6. Confusion matrices for the survey questions Q1 to Q28.

TABLE 7. Selected features for each target class attribute.

ID	Target Class Attribute	Selected Features	ID	Target Class Attribute	Selected Features
1	instr	repeat, class, Q18	17	Q13	Q14, Q16, Q15
2	repeat	attendance, difficulty, instr	18	Q14	Q,13 Q15, Q17
3	attendance	difficulty, repeat, Q17	19	Q15	Q13, Q14, Q17
4	difficulty	attendance, repeat, Q17	20	Q16	Q13, Q15, Q18
5	Q1	Q2, Q4, Q5	21	Q17	Q15, Q19, Q22
6	Q2	Q1, Q3, Q4	22	Q18	Q16, Q19, Q20
7	Q3	Q2, Q,4 Q5	23	Q19	Q18, Q20, Q21
8	Q4	Q2, Q5, Q6	24	Q20	Q19, Q21, Q22
9	Q5	Q4, Q6, Q7	25	Q21	Q20, Q22, Q25
10	Q6	Q4, Q5, Q7	26	Q22	Q20, Q21, Q25
11	Q7	Q5, Q6, Q8	27	Q23	Q21, Q22, Q24
12	Q8	Q5, Q7, Q10	28	Q24	Q23, Q25, Q26
13	Q9	Q8, Q10, Q11	29	Q25	Q21, Q22, Q28
14	Q10	Q7, Q8, Q9	30	Q26	Q25, Q27, Q28
15	Q11	Q9, Q10, Q12	31	Q27	Q25, Q26, Q28
16	Q12	Q8, Q10, Q11	32	Q28	Q22, Q25, Q26

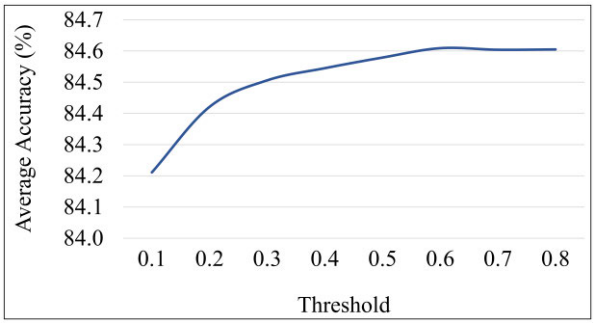


FIGURE 7. Average accuracy (%) of the RTONF model across different threshold values θ.

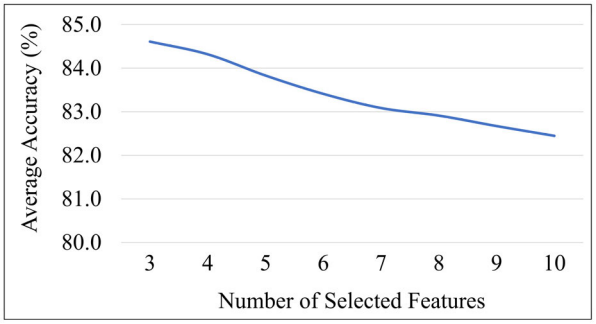


FIGURE 8. Impact of the number of selected features (f) on the average accuracy of the RTONF model.

the threshold can yield measurable gains in performance up to an optimal point. The performance is positively correlated with the parameter value up to the point $\theta = 0.6$. The plateau beyond this point suggests a diminishing return in sensitivity, where further increases in the parameter yield no performance gain. In other words, the pattern implies that the model becomes increasingly insensitive to changes in the parameter beyond the mid-range values. Therefore, careful tuning within the lower to mid-range spectrum is crucial, whereas fine-tuning at higher values may offer limited benefit. These findings imply that an intermediate threshold value effectively balances noise removal and retention of valuable information, thereby boosting the model's predictive accuracy. Based on the results, θ was set to 0.6 in subsequent experiments to reflect this empirically derived optimal setting.

The impact of varying the number of selected features (parameter f) on the average accuracy of the model is illustrated in Fig. 8. The RTONF method demonstrated a clear sensitivity to changes in f, achieving its highest accuracy at $f = 3$. This indicates that selecting a smaller subset of features can enhance generalization and reduce overfitting. As f increased from 3 to 10, the average accuracy gradually decreased, suggesting that including more features may introduce redundant or noisy information that negatively impacts the model's predictive ability. This outcome underscores the importance of selecting the most relevant features for optimal model performance. The sensitivity of the model to feature dimension implies that careful tuning of f is essential to balance model complexity and predictive accuracy.

The hyperparameter configuration of the random tree classifier was specified as follows. The k-value parameter—denoting the number of attributes randomly selected at each

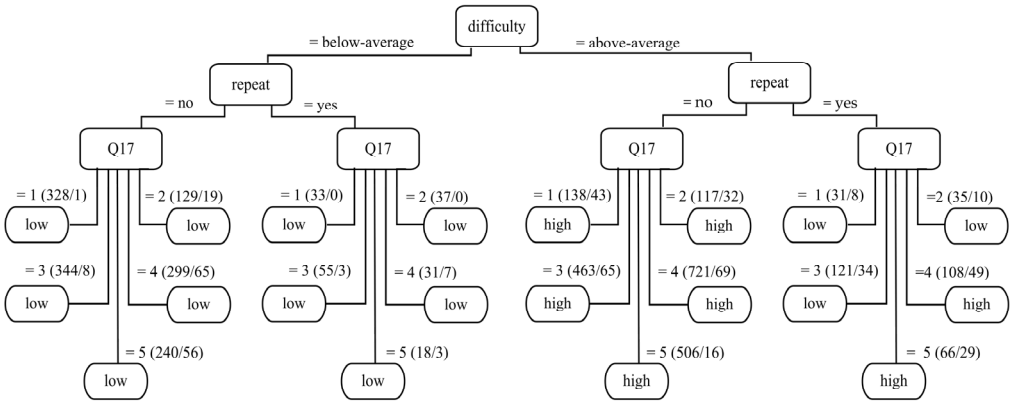


FIGURE 9. Random tree for the proposed RTONF method.

decision node—was set to 0, meaning that all available features are considered at every split. The ''minNum'' parameter, controlling the minimum number of instances required per leaf, is also set to 1, permitting highly detailed splits. The random tree relied on the Gini index to evaluate split quality. While this configuration enables exhaustive exploration of the dataset's structure, it also carries the risk of capturing noise within the training data. To address this concern, the RTONF framework integrates ordinal noise filtering and feature selection processes that effectively clean and reduce the dataset prior to training, thereby mitigating the potential for overfitting.

G. RANDOM TREE STRUCTURE

To better understand the internal decision-making process of the proposed RTONF method, it is essential to examine the structure of a typical tree generated during training. Fig. 9 illustrates a sample random tree generated during a fold of the proposed method. This tree aims to model student attendance—a crucial indicator of student satisfaction—based on survey responses, while also addressing the importance of applying and developing interpretable machine learning models. The root node splits on the ''difficulty'' attribute, which captures the degree of difficulty of the course as perceived by students. Subsequent splits occur on the ''repeat'' attribute, representing the number of times a student is taking the course—a factor that can significantly impact satisfaction and, consequently, attendance. The final splits are determined by the ordinal attribute Q17, which measures whether the instructor arrived on time for classes and is coded from 1 (strongly disagree) to 5 (strongly agree). The target variable, ''attendance'', is coded on a scale of $\{0, 1, 2, 3, 4\}$, representing increasing levels of attendance. Notably, higher satisfaction often correlates with increased attendance. The class labels at the leaves (''high'' or ''low'' attendance) reflect the predicted likelihood of a student attending the class based on these combined factors. Each leaf node is annotated with numbers in parentheses, providing information about the predicted class and how the instances are distributed at that point in the tree. This interpretable model, with a total of 27 nodes, effectively indicates critical factors influencing student attendance and underscores the RTONF's capability to extract actionable understandings from survey data while remaining transparent and easy to analyze.

VI. DISCUSSION

This section provides in-depth comparative advantages of the proposed RTONF method in the context of existing literature. The results in Table 8 clearly demonstrate the superiority of the proposed method over several state-of-the-art approaches [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74] evaluated on the same TSED dataset. The table summarizes key comparative details, including references to prior works, publication years, the methods with their reported results, as well as information on data preprocessing techniques, feature selection strategies, target class attributes, and evaluation metrics. For clarity, the following abbreviations are used in Table 8: not available (NA), cross-validation (CV), data transformation (DT), data cleaning (DC), data discretization (DD), data imbalance (DI), data normalization (DN), data sampling (DS), feature selection (FS), accuracy (ACC), and absolute error (AE). To ensure a comprehensive evaluation, the proposed method is compared with existing approaches across four distinct target class attributes, including attendance, difficulty, instructor, and repeat, to indicate the relative performance and contextual strengths of the RTONF method as follows:

- When the attendance variable was chosen as the target class attribute, the accuracies of previous state-of-the-art methods [62], [67] ranged between 45.46% and 54.60%, whereas the RTONF method demonstrated a markedly superior predictive performance with an accuracy of 71.46%. This notable improvement underscores the effectiveness of RTONF in predicting students' attendance levels as a classification task.
- When the difficulty variable was selected as the target attribute, RTONF attained an accuracy of 71.67%, outperforming previous models, including logistic regression (LR) (64.31%), support vector machines (SVM)

TABLE 8. The comparison of the proposed method (RTONF) with state-of-the-art methods on the same TSED dataset.

Ref.	Authors	Year	Preprocessing	Feature Selection	Target	Evaluation Metric	Methods	Result
[62]	Shou et al.	2025	DN, FS	RP-GA	Attendance	ACC	SVM NB MLP RF	53.10 48.00 54.60 53.60
[63]	Schumacher et al.	2025	DT, DC, DS	NA	NA	AE	EM	83.60
[64]	Chen et al.	2024	FS	Weighted fuzzy margin	NA	ACC	Weighted KNN	51.82
[61]	Chen and Wu	2024	DT, FS	User-defined	Difficulty	ACC	LR Distributed LR SVM Distributed SVM	64.31 64.09 64.88 64.93
[65]	Serapiao et al.	2023	DT, DC	NA	NA	AE	SC-SQ CPE MC-SQ FMC-SQ FMC-MQ	72.30 66.20 68.50 55.30 65.60
[66]	Gao et al.	2022	DD, FS	Three-way Approximate FS	NA	ACC	KNN SVM	52.85 62.01
[58]	Thaher et al.	2021	DT, DI, DS, FS	EWOA	Repeat	ACC	KNN DTree LDA LB NB	82.20 80.46 84.20 83.94 83.76
[67]	Xiao et al.	2021	DC, DD, FS	RnkHEU	Attendance	ACC	NB DTree (C4.5) MLP KNN	46.30 47.53 45.46 47.40
[68]	Wu et al.	2021	NA	NA	NA	Precision	SKCSE	75.05
[55]	Thaher and Jayousi	2020	DI, FS, DN	Information Gain	Repeat	ACC	MLP	57.90
[69]	Selvapandian et al.	2020	FS	Probabilistic Membership	Instructor	ACC	EF-NN K-Mean	93.00 90.00
[70]	Afrin et al.	2020	FS	Mutual Information	Q8 Q9 Q10 Q11 Q12	ACC	SVM	84.16 80.69 84.02 80.82 79.86
[56]	Li et al.	2019	DT, DI	NA	Repeat	G-Mean	Improved RVFL Networks	72.76
[71]	Zhu and Miao	2019	FS	Variable Features	NA	ACC	SVM FDROP KARMA CNN SOMVFV	74.85 79.08 75.29 77.20 79.13
[72]	Thakar et al.	2018	DS, FS	Chi-Square	Repeat	ACC	RPM PCA	83.30 82.11
[57]	Can et al.	2018	DT, FS	User-defined	Repeat	ACC	LR	84.11
[60]	Ahmed et al.	2016	DT, FS	OneR	Instructor	ACC	DTree (J48) NB SVM MLP	85.10 84.30 85.80 84.60
[73]	Boriratrit et al.	2016	NA	NA	NA	ACC	Harmonic ELM Unsupervised ELM K-Harmonic Mean K-Mean	40.72 42.14 37.78 38.21
[59]	Hajizadeh and Ahmadzadeh	2014	DT, FS	User-defined	Repeat	ACC	REPTree	84.26
[74]	Fokoue and Gunduz	2013	DT, DD	User-defined	Instructor	ACC	DTree	93.00
	Proposed Method		DT, FS	Pearson correlation coefficient	Instructor Repeat Attendance Difficulty	ACC	RTONF	98.26 84.35 71.46 71.67

TABLE 8. *(Continued.)* **The comparison of the proposed method (RTONF) with state-of-the-art methods on the same TSED dataset.**

	Q8	83.38
	Q9	82.22
	Q10	83.37
	Q11	82.13
	Q12	82.20

CPE = Class-prevalence ensemble, EF-NN = Efficient fusion-based neural network, ELM = Extreme learning machine, EM = Expectation-maximization, EWOA = Enhanced whale optimization algorithm, FDROP = Feature dropout, FMC-MQ = Fusioned multiple classifiers with multiple quantifiers, FMC-SQ = Fusioned multiple classifiers with single quantifier, KARMA =Kernelized algorithm for risk-minimization with missing attributes, LB = LogitBoost,, LDA = Linear discriminant analysis,, MC-SQ = Multiple-classifier with single-quantifier, PCA = Principal component analysis, RnkHEU = Rank and heuristic feature selection, RP-GA = Random partial-genetic algorithm, RPM = Robust prediction model, RVFL = Random vector functional-link, SC-SQ = Single classifier with single quantifier, SKCSE = Semi-supervised k-means clustering with selective ensemble, SOMVFV = Semi-supervised one-pass multiview learning with variable features and views

(64.88%), and their distributed versions (64.09% and 64.93%, respectively) [58]. These improvements point out RTONF's capacity to model complex decision boundaries, making it particularly suitable for applications in educational data mining.

- When the instructor variable was designated as the target attribute, RTONF achieved significant improvements over widely used baseline methods [60], [69], [74]. For instance, RTONF reached an accuracy of 98.26%, outperforming J48 at 85.10%, naive Bayes (NB) (84.30%), support vector machine (SVM) (85.80%), and multi-layer perceptron (MLP) (84.60%) [60]. As another example, RTONF delivered a 5.7% accuracy gain over the previously proposed efficient fusion-based neural network (EF-NN) [69]. A similar pattern was observed in another study [74], where RTONF exceeded the performance of a prior decision tree-based method by the same margin. These findings collectively reveal the improved classification capability of the proposed method over the existing approaches.

- When the repeat variable was assigned as the target attribute, our proposed approach demonstrated superior predictive accuracy (84.35%) compared to all previous models [55], [56], [57], [58], [59], and [72], including robust prediction model (RPM) (83.30%) and principal component analysis (PCA) (82.11%) [72]. For instance, RTONF represented a clear improvement over the k-nearest neighbors (KNN) algorithm, which achieved only 82.20% accuracy. This result highlights our method's improved capability to model complex relationships in educational data, in contrast to the relatively simplistic distance-based mechanism employed by KNN [58]. Compared to the naive Bayes (NB) model [58], which reached an accuracy of 83.76%, our approach exhibited a modest yet meaningful enhancement in performance. RTONF also outperformed other tree-based classifiers [58], [60]; for instance, the reduced error pruning tree (REPTree) recorded a slightly lower accuracy of 84.26% [59], while a standard decision tree model reported 80.46% [58]. Furthermore, LogitBoost (LB) [58] yielded an accuracy of 83.94%, still falling short of our model's performance. This indicates that RTONF can offer an edge in performance even over ensemble learning methods. The largest margin of improvement was observed against multi-layer perceptron (MLP) [55], confirming the strength of RTONF.
- Random Tree was selected as the base classifier due to its simplicity, interpretability, and responsiveness to the preprocessing step, making it a suitable choice for the focus of this study. It provides a favorable balance between accuracy, simplicity, and computational efficiency. In contrast, models such as SVM and neural networks generally require higher computational cost, which makes them less suitable as base learners in the scope of this work. In particular, to ensure transparency and facilitate explainable artificial intelligence (xAI)-related interpretation, we focused on tree-based algorithms during the tests. To assess the broader applicability of the proposed ordinal noise filtering and feature selection method (ONF), other commonly used classifiers, C4.5 and RepTree were also tested. The findings showed that the proposed preprocessing step improved classification accuracy from 83.94% (before ONF) to 84.57% (after ONF) for C4.5, and from 84.01% (before ONF) to 84.46% (after ONF) for RepTree. Although the effect varies across models, the overall trend indicates that ONF is beneficial, with the most pronounced improvement observed in Random Tree. These gains indicate that the proposed method effectively reduces the negative impact of noisy instances across different algorithms.

Collectively, all the comparisons reflect the precision and consistency of our method across a wide range of ML models, reinforcing its applicability for student survey data analysis in the educational domain. To further substantiate the reliability of the comparative improvements, statistical significance tests were performed. The Wilcoxon signed-rank test yielded a p-value of 7.55×10^{-7}, and the Mann–Whitney U test produced a p-value of 0.014. Both results are well below the significance threshold of 0.05, thereby approving that

the performance gains achieved by the proposed RTONF method over the state-of-the-art approaches are statistically significant rather than attributable to random variation. These findings provide strong empirical evidence that the proposed method consistently delivers superior predictive performance across different target class attributes.

VII. CONCLUSION AND FUTURE WORK

This study proposes an innovative hybrid machine learning approach, named RTONF, which integrates a tree-based algorithm with ordinal noise detection and feature selection for student survey analysis. The method is specifically designed to predict both student performance and student satisfaction, providing a comprehensive understanding of educational outcomes that go beyond a single aspect of student evaluation. Utilizing the real-world TSED dataset, which includes comprehensive student evaluations using a Likert-type scale (poor < fair < good < very good < excellent), the method acknowledges the inherent order of class labels—an aspect often overlooked in prior works—thus ensuring that the noise identification process is both meaningful and relevant to the educational context. By respecting this ordered structure, RTONF effectively mitigates the risk of mislabeling, which can otherwise lead to biased predictions and diminished model performance.

Furthermore, the use of Pearson Correlation for feature selection helps identify the most informative attributes to increase computational efficiency while improving classification accuracy. This step is critical for educational stakeholders who require clear explanations of the factors influencing student performance and satisfaction. We developed interpretable machine learning models that promote transparency via random trees as the core classifiers with an average accuracy of 84.61%, precision of 84.67%, recall of 84.61%, and F-measure of 84.64%. The empirical evaluation revealed substantial gains across multiple performance metrics when compared to the traditional random tree classifier, including an improvement of 5.23% in accuracy, 5.43% in precision, 5.23% in recall, and 5.33% in F-measure, underlining the efficacy of the proposed solution. These results confirm that by combining ordinal noise filtering, effective feature selection, and a robust classification algorithm, RTONF successfully addresses the challenges posed by noisy survey data and supports more actionable understandings for educational decision-making.

While the findings are encouraging, one limitation of this study is that they are derived from a single real-world dataset. Future research will therefore focus on validating the framework across multiple ordinal datasets and diverse domains to provide stronger evidence of its generalizability. It is also worth noting that some methodological constraints should be acknowledged. For instance, class imbalance in ordinal data can affect the stability of the method's performance and may require additional resampling or weighting strategies. Furthermore, the sensitivity of the results to parameter settings could influence reproducibility across diverse datasets. Additionally, the computational costs associated with the method's implementation may limit scalability in large-scale applications. Future research will aim to mitigate these constraints by exploring advanced imbalance-handling techniques, automated hyperparameter tuning, and more efficient optimization strategies.

Building on the strong performance of the RTONF method, further works could also explore integrating this approach into educational analytics platforms that support timely perceptions, enabling dynamic feedback and adaptive learning pathways. Such integration could empower educators and administrators with immediate, data-informed decision-making tools. Exploring the application of RTONF across different educational contexts or extending it to mobile and cloud-based platforms could further boost its reach and usability. These developments would not only broaden the impact of RTONF but also emphasize its versatility in addressing a wide range of challenges within the educational domain and contributing to improved student outcomes.

REFERENCES

[1] B. T. M. Wong, K. C. Li, and M. Liu, "The role of learning analytics in evaluating course effectiveness," *Sustainability*, vol. 17, no. 2, p. 559, Jan. 2025, doi: 10.3390/su17020559.

[2] M. Yağcı, "Educational data mining: Prediction of students' academic performance using machine learning algorithms," *Smart Learn. Environments*, vol. 9, no. 1, pp. 1–19, Dec. 2022, doi: 10.1186/s40561-022-00192-z.

[3] S. Gupta and A. Gupta, "Dealing with noise problem in machine learning data-sets: A systematic review," *Proc. Comput. Sci.*, vol. 161, pp. 466–474, Jul. 2019, doi: 10.1016/j.procs.2019.11.146.

[4] J. Qiu, Q. Wu, G. Ding, Y. Xu, and S. Feng, "A survey of machine learning for big data processing," *EURASIP J. Adv. Signal Process.*, vol. 2016, no. 1, pp. 1–16, Dec. 2016, doi: 10.1186/s13634-016-0355-x.

[5] E. Frank and R. Kirkby. *Random Tree*. Accessed: May 2, 2025. [Online]. Available: http://weka.sourceforge.net/doc.dev/weka/classifiers/trees/RandomTree.html

[6] S. Li, T. Wang, H. Yin, S. Ding, and Z. Cai, "Behavioral analysis of postgraduate education satisfaction: Unveiling key influencing factors with Bayesian networks and feature importance," *Behav. Sci.*, vol. 15, no. 4, p. 559, Apr. 2025, doi: 10.3390/bs15040559.

[7] S. Albahli, "Advancing sustainable educational practices through AI-driven prediction of academic outcomes," *Sustainability*, vol. 17, no. 3, p. 1087, Jan. 2025, doi: 10.3390/su17031087.

[8] L. Zhang, J. Gao, L. Zhao, Z. Liu, and A. Guan, "Predicting college student engagement in physical education classes using machine learning and structural equation modeling," *Appl. Sci.*, vol. 15, no. 7, p. 3884, Apr. 2025, doi: 10.3390/app15073884.

[9] M. A. Aslam, F. Murtaza, M. E. Ul Haq, A. Yasin, and N. Ali, "SAPEx-D: A comprehensive dataset for predictive analytics in personalized education using machine learning," *Data*, vol. 10, no. 3, p. 27, Feb. 2025, doi: 10.3390/data10030027.

[10] G. Karalekas, S. Vologiannidis, and J. Kalomiros, "Teaching artificial intelligence and machine learning in secondary education: A robotics-based approach," *Appl. Sci.*, vol. 15, no. 8, p. 4570, Apr. 2025, doi: 10.3390/app15084570.

[11] F. E. Terrazas-Arellanes, L. Strycker, G. G. Alvez, B. Miller, and K. Vargas, "Promoting agency among upper elementary school teachers and students with an artificial intelligence machine learning system to score performance-based science assessments," *Educ. Sci.*, vol. 15, no. 1, p. 54, Jan. 2025, doi: 10.3390/educsci15010054.

[12] A. Angeioplastis, J. Aliprantis, M. Konstantakis, and A. Tsimpiris, "Predicting student performance and enhancing learning outcomes: A data-driven approach using educational data mining techniques," *Computers*, vol. 14, no. 3, p. 83, Feb. 2025, doi: 10.3390/computers14030083.

[13] G. Mimić, A. K. Mishra, M. Marković, B. Živaljević, D. Pavlović, and O. Marko, "Machine learning-based harvest date detection and prediction using SAR data for the vojvodina region (Serbia)," *Sensors*, vol. 25, no. 7, p. 2239, Apr. 2025, doi: 10.3390/s25072239.

[14] J. Wang and Y. Yu, "Machine learning approach to student performance prediction of online learning," *PLoS ONE*, vol. 20, no. 1, Jan. 2025, Art. no. e0299018, doi: 10.1371/journal.pone.0299018.

[15] N. Mirbahar, K. Kumar, and A. Ali Laghari, "Enhancing mobile app recommendations with crowdsourced educational data using machine learning and deep learning," *IEEE Access*, vol. 13, pp. 38778–38792, 2025, doi: 10.1109/ACCESS.2025.3541116.

[16] I. Gómez-Talal, L. Bote-Curiel, and J. L. Rojo-Álvarez, "Interpretable machine learning models for PISA results in mathematics," *IEEE Access*, vol. 13, pp. 27371–27397, 2025, doi: 10.1109/ACCESS.2025.3538585.

[17] M. A. López-Cueva, S. J. Zegarra-Ugarte, E. H. Torres-Pacheco, R. F. Arpasi-Chura, P. Enríquez-Salas, and P. M. López-Paz, "Prediction of university student satisfaction in the educational experience using decision trees and logistic regression: A comparative analysis," *Pakistan J. Life Social Sci. (PJLSS)*, vol. 22, no. 2, pp. 7844–7856, 2024, doi: 10.57239/pjlss-2024-22.2.00591.

[18] T. Zhang, Z. Zhong, W. Mao, Z. Zhang, and Z. Li, "A new machine-learning-driven grade-point average prediction approach for college students incorporating psychological evaluations in the post-COVID-19 era," *Electronics*, vol. 13, no. 10, p. 1928, May 2024, doi: 10.3390/electronics13101928.

[19] M. A. Hassan, A. H. Muse, and S. Nadarajah, "Predicting student dropout rates using supervised machine learning: Insights from the 2022 national education accessibility survey in Somaliland," *Appl. Sci.*, vol. 14, no. 17, p. 7593, Aug. 2024, doi: 10.3390/app14177593.

[20] J. P. J. Pires, F. Brito Correia, A. Gomes, A. R. Borges, and J. Bernardino, "Predicting student performance in introductory programming courses," *Computers*, vol. 13, no. 9, p. 219, Sep. 2024, doi: 10.3390/computers13090219.

[21] M. M. Ncube and P. Ngulube, "Optimising data analytics to enhance postgraduate student academic achievement: A systematic review," *Educ. Sci.*, vol. 14, no. 11, p. 1263, Nov. 2024, doi: 10.3390/educsci14111263.

[22] M. A. Rahman and T. Kohli, "Mental health analysis of international students using machine learning techniques," *PLoS ONE*, vol. 19, no. 6, Jun. 2024, Art. no. e0304132, doi: 10.1371/journal.pone.0304132.

[23] S. Mohd Zaki, S. Razali, M. A. R. Awang Kader, M. Z. Laton, M. Ishak, and N. Mohd Burhan, "Predicting students' performance at higher education institutions using a machine learning approach," *Kybernetes*, early access, Aug. 2024, doi: 10.1108/k-12-2023-2742.

[24] H. Xu and M. Kim, "Combination prediction method of students' performance based on ant colony algorithm," *PLoS ONE*, vol. 19, no. 3, Mar. 2024, Art. no. e0300010, doi: 10.1371/journal.pone.0300010.

[25] N. R. Raji, R. M. S. Kumar, and C. L. Biji, "Explainable machine learning prediction for the academic performance of deaf scholars," *IEEE Access*, vol. 12, pp. 23595–23612, 2024, doi: 10.1109/ACCESS.2024.3363634.

[26] Z. Pan and M. Cutumisu, "Using machine learning to predict U.K. and Japanese secondary students' life satisfaction in PISA 2018," *Brit. J. Educ. Psychol.*, vol. 94, no. 2, pp. 474–498, Jun. 2024, doi: 10.1111/bjep.12657.

[27] A. M. Mutawa and S. Sruthi, "Enhancing human–computer interaction in online education: A machine learning approach to predicting student emotion and satisfaction," *Int. J. Human–Computer Interact.*, vol. 40, no. 24, pp. 8827–8843, Dec. 2024, doi: 10.1080/10447318.2023.2291611.

[28] J. de Souza Zanirato Maia, A. P. A. Bueno, and J. R. Sato, "Applications of artificial intelligence models in educational analytics and decision making: A systematic review," *World*, vol. 4, no. 2, pp. 288–313, May 2023, doi: 10.3390/world4020019.

[29] W. Villegas-Ch, J. Govea, and S. Revelo-Tapia, "Improving student retention in institutions of higher education through machine learning: A sustainable approach," *Sustainability*, vol. 15, no. 19, p. 14512, Oct. 2023, doi: 10.3390/su151914512.

[30] E. G. Bayirli, A. Kaygun, and E. Öz, "An analysis of PISA 2018 mathematics assessment for asia-pacific countries using educational data mining," *Mathematics*, vol. 11, no. 6, p. 1318, Mar. 2023, doi: 10.3390/math11061318.

[31] B. Holicza and A. Kiss, "Predicting and comparing students' online and offline academic performance using machine learning algorithms," *Behav. Sci.*, vol. 13, no. 4, p. 289, 2023, doi: 10.3390/bs13040289.

[32] R. V. Anand, A. Q. Md, S. Urooj, S. Mohan, M. A. Alawad, and A. C., "Enhancing diagnostic decision-making: Ensemble learning techniques for reliable stress level classification," *Diagnostics*, vol. 13, no. 22, p. 3455, Nov. 2023, doi: 10.3390/diagnostics13223455.

[33] N. Alruwais and M. Zakariah, "Evaluating student knowledge assessment using machine learning techniques," *Sustainability*, vol. 15, no. 7, p. 6229, Apr. 2023, doi: 10.3390/su15076229.

[34] N. Alruwais and M. Zakariah, "Student-engagement detection in classroom using machine learning algorithm," *Electronics*, vol. 12, no. 3, p. 731, Feb. 2023, doi: 10.3390/electronics12030731.

[35] R. Nawaz, Q. Sun, M. Shardlow, G. Kontonatsios, N. R. Aljohani, A. Visvizi, and S.-U. Hassan, "Leveraging AI and machine learning for national student survey: Actionable insights from textual feedback to enhance quality of teaching and learning in U.K.'s higher education," *Appl. Sci.*, vol. 12, no. 1, p. 514, Jan. 2022, doi: 10.3390/app12010514.

[36] E. Abu-Taieh, I. AlHadid, R. Masa'deh, R. S. Alkhawaldeh, S. Khwaldeh, and A. Alrowwad, "Factors influencing YouTube as a learning tool and its influence on academic achievement in a bilingual environment using extended information adoption model (IAM) with ML prediction—Jordan case study," *Appl. Sci.*, vol. 12, no. 12, p. 5856, Jun. 2022, doi: 10.3390/app12125856.

[37] L. Falát and T. Piscová, "Predicting GPA of university students with supervised regression machine learning models," *Appl. Sci.*, vol. 12, no. 17, p. 8403, Aug. 2022, doi: 10.3390/app12178403.

[38] Y. A. Alsariera, Y. Baashar, G. Alkawsi, A. Mustafa, A. A. Alkahtani, and N. Ali, "Assessment and evaluation of different machine learning algorithms for predicting student performance," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–11, May 2022, doi: 10.1155/2022/4151487.

[39] M. Jamjoom, E. Alabdulkreem, M. Hadjouni, F. Karim, and M. Qarh, "Early prediction for at-risk students in an introductory programming course based on student self-efficacy," *Informatica*, vol. 45, no. 6, pp. 1–9, 2021, doi: 10.31449/inf.v45i6.3528.

[40] K. Nahar, B. I. Shova, T. Ria, H. B. Rashid, and A. H. M. S. Islam, "Mining educational data to predict students performance," *Educ. Inf. Technol.*, vol. 26, no. 5, pp. 6051–6067, Sep. 2021, doi: 10.1007/s10639-021-10575-3.

[41] J. Kabathova and M. Drlik, "Towards predicting student's dropout in university courses using different machine learning techniques," *Appl. Sci.*, vol. 11, no. 7, p. 3130, Apr. 2021, doi: 10.3390/app11073130.

[42] I. M. K. Ho, K. Y. Cheong, and A. Weldon, "Predicting student satisfaction of emergency remote learning in higher education during COVID-19 using machine learning techniques," *PLoS ONE*, vol. 16, no. 4, Apr. 2021, Art. no. e0249423, doi: 10.1371/journal.pone.0249423.

[43] J. M. Barrios and P. E. Romero, "Decision tree methods for predicting surface roughness in fused deposition modeling parts," *Materials*, vol. 12, no. 16, p. 2574, Aug. 2019, doi: 10.3390/ma12162574.

[44] S. Kumar, C. S. Prasad Ojha, N. K. Tiwari, and S. Ranjan, "Exploring the potential of artificial intelligence techniques in prediction of the removal efficiency of vortex tube silt ejector," *Int. J. Sediment Res.*, vol. 38, no. 4, pp. 615–627, Aug. 2023, doi: 10.1016/j.ijsrc.2023.03.001.

[45] S. O. Giwa, R. T. Taziwa, and M. Sharifpur, "Dependence of composition-based approaches on hybrid biodiesel fuel properties prediction using artificial neural network and random tree algorithms," *Renew. Energy*, vol. 218, Dec. 2023, Art. no. 119324, doi: 10.1016/j.renene.2023.119324.

[46] M. Kumar, A. Elbeltagi, C. B. Pande, A. N. Ahmed, M. F. Chow, Q. B. Pham, A. Kumari, and D. Kumar, "Applications of data-driven models for daily discharge estimation based on different input combinations," *Water Resour. Manage.*, vol. 36, no. 7, pp. 2201–2221, May 2022, doi: 10.1007/s11269-022-03136-x.

[47] U. Fareed, U. Khadam, M. M. Iqbal, and M. J. Iqbal, "Road accidents investigation and forecasting using data mining techniques," *KIET J. Comput. Inf. Sci.*, vol. 6, pp. 28–49, Aug. 2022.

[48] M. Goga, S. Kuyoro, and N. Goga, "A recommender for improving the student academic performance," *Proc. Social Behav. Sci.*, vol. 180, pp. 1481–1488, May 2015, doi: 10.1016/j.sbspro.2015.02.296.

[49] G. Jiang, F. Wang, and W. Wang, "Noise cleaning for nonuniform ordinal labels based on inter-class distance," *Appl. Intell.*, vol. 54, no. 11, pp. 6997–7011, 2024, doi: 10.1007/s10489-024-05551-6.

[50] M. Iannario and A. C. Monti, "Generalized residuals and outlier detection for ordinal data with challenging data structures," *Stat. Methods Appl.*, vol. 32, no. 4, pp. 1197–1216, Oct. 2023, doi: 10.1007/s10260-023-00686-1.

[51] G. Jiang, W. Wang, Y. Qian, and J. Liang, "A unified sample selection framework for output noise filtering: An error-bound perspective," *J. Mach. Learn. Res.*, vol. 22, no. 18, pp. 1–66, 2021.

[52] E. Frank, M. A. Hall, and I. H. Witten, *The WEKA Workbench; Online Appendix for Data Mining: Practical Machine Learning Tools and Techniques*, 4th ed., San Mateo, CA, USA: Morgan Kaufmann, 2016.

[53] H. S. Daliri, M. Dejagah, H. R. Baghaee, H. A. Abyaneh, and A. Bakhshai, "HMM-based feature extraction and machine learning methods for event detection and classification in microgrids," *IEEE Trans. Smart Grid*, early access, Jul. 18, 2025, doi: 10.1109/TSG.2025.3590518.

[54] E. Fokoue and N. Gunduz. (2013). *Turkiye Student Evaluation*. UCI Machine Learning Repository. Accessed: May 2, 2025. [Online]. Available: https://archive.ics.uci.edu/dataset/262/turkiye

[55] T. Thaher and R. Jayousi, "Prediction of student's academic performance using feedforward neural network augmented with stochastic trainers," in *Proc. IEEE 14th Int. Conf. Appl. Inf. Commun. Technol. (AICT)*, Tashkent, Uzbekistan, Oct. 2020, pp. 1–7.

[56] M. Li, C. Huang, D. Wang, Q. Hu, J. Zhu, and Y. Tang, "Improved randomized learning algorithms for imbalanced and noisy educational data classification," *Computing*, vol. 101, no. 6, pp. 571–585, Jun. 2019, doi: 10.1007/s00607-018-00698-w.

[57] Ş. Can, T. Özdil, and C. Yılmaz, "Üniversite öğrencilerinin ders başarisini etkileyen faktörlerin lojistik regresyon analizi ıle tahminlenmesi," *Int. Rev. Econ. Manage.*, vol. 6, no. 1, pp. 28–49, Apr. 2018, doi: 10.18825/iremjournal.349984.

[58] T. Thaher, A. Zaguia, S. Al Azwari, M. Mafarja, H. Chantar, A. Abuhamdah, H. Turabieh, S. Mirjalili, and A. Sheta, "An enhanced evolutionary student performance prediction model using whale optimization algorithm boosted with sine-cosine mechanism," *Appl. Sci.*, vol. 11, no. 21, p. 10237, Nov. 2021, doi: 10.3390/app112110237.

[59] N. Hajizadeh and M. Ahmadzadeh, "Analysis of factors that affect the students academic performance–data mining approach," 2014, *arXiv:1409.2222*.

[60] A. M. Ahmed, A. Rizaner, and A. H. Ulusoy, "Using data mining to predict instructor performance," *Procedia Comput. Sci.*, vol. 102, pp. 137–142, Aug. 2016, doi: 10.1016/j.procs.2016.09.380.

[61] M. Y. Chen and Q. Wu, "Distributed classification by divide and conquer approach," in *Proc. IEEE 3rd Int. Conf. Comput. Mach. Intell. (ICMI)*, Mt Pleasant, MI, USA, Apr. 2024, pp. 1–5, doi: 10.1109/icmi60790.2024.10586091.

[62] Z. Shou, Z. Zhang, J. Chen, H. Yuan, and J. Mo, "All element selection method in classroom social networks and analysis of structural characteristics," *Int. J. Adv. Comput. Sci. Appl.*, vol. 16, no. 2, pp. 492–503, 2025, doi: 10.14569/ijacsa.2025.0160251.

[63] T. Schumacher, M. Strohmaier, and F. Lemmerich, "A comparative evaluation of quantification methods," *J. Mach. Learn. Res.*, pp. 1–54, 2021. [Online]. Available: http://jmlr.org/papers/v26/21-0241.html

[64] Z. Chen, G. Chen, C. Gao, J. Zhou, and J. Wen, "Robust weighted fuzzy margin-based feature selection with three-way decision," *Int. J. Approx. Reasoning*, vol. 173, Oct. 2024, Art. no. 109253, doi: 10.1016/j.ijar.2024.109253.

[65] A. B. S. Serapião, Z. Donyavi, and G. Batista, "Ensembles of classifiers and quantifiers with data fusion for quantification learning," in *Proc. Discovery Sci. (DS)* (Lecture Notes in Computer Science), vol. 14276, A. Bifet, A. C. Lorena, R. P. Ribeiro, J. Gama, P. H. Abreu, Eds., Cham, Switzerland: Springer, Oct. 2023, pp. 1–15, doi: 10.1007/978-3-031-45275-8_1.

[66] C. Gao, Z. Wang, and J. Zhou, "Three-way approximate reduct based on information-theoretic measure," *Int. J. Approx. Reasoning*, vol. 142, pp. 324–337, Mar. 2022, doi: 10.1016/j.ijar.2021.12.008.

[67] W. Xiao, P. Ji, and J. Hu, "RnkHEU: A hybrid feature selection method for predicting students' performance," *Sci. Program.*, vol. 2021, pp. 1–16, 2021, doi: 10.1155/2021/1670593.

[68] X. Wu, C. Guo, and T. Hu, "The research on clustering ensembles selection algorithm based on semi-supervised K-means clustering," *J. Phys., Conf. Ser.*, vol. 1732, no. 1, Jan. 2021, Art. no. 012074, doi: 10.1088/1742-6596/1732/1/012074.

[69] D. Selvapandian, T. W. Meshach, K. S. S. Babu, R. Dhanapal, and J. D. Immanuel, "An efficient sentiment analysis on feedback assessment from student to provide better education," in *Proc. 4th Int. Conf. I-SMAC (IoT Social, Mobile, Analytics Cloud) (I-SMAC)*, Palladam, India, Oct. 2020, pp. 1293–1300, doi: 10.1109/I-SMAC49090.2020.9243594.

[70] F. Afrin, M. S. Rahaman, and M. Hamilton, "Mining student responses to infer student satisfaction predictors," 2020, *arXiv:2006.07860*.

[71] C. Zhu and D. Miao, "Semi-supervised one-pass multi-view learning with variable features and views," *Neural Process. Lett.*, vol. 50, no. 1, pp. 189–226, 2019, doi: 10.1007/s11063-019-10037-5.

[72] P. Thakar, A. Mehta, and Manisha, "Robust prediction model for multidimensional and unbalanced datasets," 2024, *arXiv:2406.03507*.

[73] S. Boriratrit, S. Chiewchanwattana, K. Sunat, P. Musikawan, and P. Horata, "Harmonic extreme learning machine for data clustering," in *Proc. 13th Int. Joint Conf. Comput. Sci. Softw. Eng. (JCSSE)*, Khon Kaen, Thailand, Jul. 2016, pp. 1–5, doi: 10.1109/JCSSE.2016.7748872.

[74] E. Fokoue and N. Gündüz. (2013). *Data Mining and Machine Learning Techniques for Extracting Patterns in Students' Evaluations of Instructors*. RIT Scholar Works. [Online]. Available: https://repository.rit.edu/article/1746



GOKSU TUYSUZOGLU was born in Giresun, Türkiye, in 1990. She received the B.S. degree in information systems engineering (double major in industrial engineering) from Dogus University, Türkiye, in 2013, the M.S. degree in computer engineering from İstanbul Technical University, Türkiye, in 2016, and the Ph.D. degree in computer engineering from Dokuz Eylul University, Türkiye, in 2021. From 2014 to 2016, she was a Research and Teaching Assistant with the Department of Computer Engineering, İstanbul Technical University. She was a Research and Teaching Assistant with the Department of Computer Engineering, Dokuz Eylul University, from 2016 to 2023, where she has been an Assistant Professor, since 2023, focusing her research on machine learning and data mining.



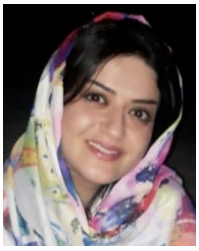
YUNUS DOGAN was born in İzmir, Türkiye, in 1985. He received the B.S., M.S., and Ph.D. degrees in computer engineering from Dokuz Eylul University, Türkiye, in 2007, 2009, and 2015, respectively. From 2007 to 2015, he was a Research Assistant with the Faculty of Engineering, Dokuz Eylul University. Since 2021, he has been an Assistant Professor with Dokuz Eylul University, focusing his research on machine learning, data mining, and deep learning to propose novel algorithms and approaches, to obtain up-to-date medical and marine detections, and to process different data types, such as texts and images. His research interests include outlier detection with machine learning on networks and data mining and knowledge discovery in medical information systems.



FERISTAH DALKILIC was born in Aydin, Türkiye, in 1984. She received the B.S., M.S., and Ph.D. degrees in computer engineering from Dokuz Eylul University, Türkiye, in 2006, 2009, and 2015, respectively. From 2009 to 2021, she was a Research Assistant with the Department of Computer Engineering, Dokuz Eylul University. Since 2021, she has been an Assistant Professor with the Department of Computer Engineering, Dokuz Eylul University. She has published over 40 research papers in various journals, conferences, and books. Her research interests include data mining, machine learning, naturel languages processing, graph theory, and bioinformatics.



ELIFE OZTURK KIYAK was born in Osmaniye, Türkiye, in 1989. She received the B.Sc. and M.Sc. degrees in computer engineering from İzmir Institute of Technology, İzmir, Türkiye, in 2013 and 2018, respectively, and the Ph.D. degree in computer engineering from Dokuz Eylul University, İzmir, in 2023. From 2015 to 2018, she was a Research Assistant with İzmir Institute of Technology, and Dokuz Eylul University, from 2018 to 2023. Since 2023, she has been an Independent Researcher based in İzmir. Her research interests include machine learning, data mining, artificial intelligence, and software engineering.



BITA GHASEMKHANI was born in Tabriz, Iran, in 1985. She received the B.S. degree in computer software engineering from UCNA University, Tabriz, in 2007, and the M.S. degree in computer engineering from Islamic Azad University, Tabriz, in 2010. She is currently pursuing the Ph.D. degree in computer engineering with Dokuz Eylul University, İzmir, Türkiye. From 2008 to 2010, she was a Teaching Assistant with the Department of Computer Engineering, Islamic Azad University, and an Associate Professor, from 2010 to 2022. Her professional experience also includes roles as a Data Scientist and a Project Advisor, specializing in machine learning and data mining technologies in both academic and industrial contexts. During the Ph.D. studies, she has authored more than 15 high-impact journal articles and has been invited as a keynote speaker at various national and international conferences. She is fluent in five major world languages. Throughout her academic journey, she has been consistently recognized as an outstanding student and an educator. In 2018, she received the Outstanding Educator Award (OEA), conferred across Iranian universities based on a comprehensive evaluation of academic excellence. In 2024, she received the Türkiye Success Scholarship by Türkiye in recognition of her exceptional performance as an international Ph.D. student. Her extensive academic and professional background underscores her commitment to innovation and advancement in the field. Beyond her accomplishments, her lifelong mission is to contribute meaningfully to global well-being through science and education.



KOKTEN ULAS BIRANT received the B.Sc., M.Sc., and Ph.D. degrees in computer engineering from Dokuz Eylul University, Türkiye, in 2000, 2002, and 2006, respectively. He is currently an Associate Professor with Dokuz Eylul University. He has been involved in more than 18 long-term research and development projects on computer science. He has published over 70 research papers in various journals, conferences, and books. His research interests include software engineering, artificial intelligence, and machine learning.



DERYA BIRANT (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in computer engineering from Dokuz Eylul University, Türkiye, in 2000, 2002, and 2006, respectively. Since 2023, she has been a Professor with the Computer Engineering Department, Dokuz Eylul University. She is the Head of the Computer Engineering Department. She was a Visiting Lecturer with the South East European University, in 2006, and Ege University, from 2010 to 2012. She has been involved in more than 20 long-term interdisciplinary research and development projects on data science. She has published over 140 research papers in journals, conferences, and books. Her research interests include machine learning and artificial intelligence. She has several Most Downloaded Article certifications and has graduation awards, ranking second in the Faculty and the Department. She was a recipient of the Outstanding Achievement Award, in 2010.